



Initial Environmental Examination

PUBLIC

February 2026

India: Tamil Nadu Urban Flagship Investment Program- Tranche 2

Subproject: Dedicated Water Supply Scheme for Madurai Municipal Corporation from Mullai Periyar River at Lower Camp
Part 2 of 3: Annexures (Continued)

Prepared by Madurai City Municipal Corporation for the Asian Development Bank (ADB). This is an updated version of the draft Initial Environmental Examination originally posted in July 2019 available on <https://www.adb.org/projects/documents/ind-49107-005-iee-0>.

Asian Development Bank

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Appendix 13

Map Showing the Distance from Forest Area and Proposed Project



Appendix 14

Existing Water Supply System

1. Water Supply and Sewerage System: The Madurai Municipal Corporation Water Supply is vested with the responsibility of

- (i) Promoting and securing the planned development of water supply and sewerage services, creation, operation and maintenance of the needed infrastructure and
- (ii) Implementation of perspective plans to meet both current and future requirements in the areas falling under the city limit.

2. Sources of Water for Madurai City. The Madurai Municipal Corporation supplies water to the Core city from Vaigai Dam and sub- surface water supply schemes on Vaigai River.

Table A14.1: Treatment Capacity of Water Treatment Plant

S. No.	Name of Source/Scheme	Type of Source	Present Supply (MLD)	Remarks
Scheme-I				
1	Kochadai	Infiltration Galleries	8.46	Sub-Surface Water/Ground Water
2	Kochadai	Collector Well		Not Functioning
3	Thachampathu Melakkal WSS	Infiltration Galleries	14	Sub-Surface Water/Ground Water
Scheme-II				
4	Manalur and Thiruppuvanam	Collector Wells	7	Sub-Surface Water/Ground Water
Vaigai Water Supply Scheme				
5	Vaigai WSS, Line-I	Intake Well Vaigai Dam	68	Surface Water
6	Vaigai WSS, Line-II	Intake Well Vaigai Dam	47	Surface Water
Scheme-III				
7	Vaigai River bed Sources	Infiltration wells	17.54	Sub-Surface Water/Ground Water
Scheme-IV				
8	Melur CWSS	Collector Wells in River Cauvery	30	Sub-Surface Water/Ground Water
	Total		192	--

Table A14.2: Storage Capacity of Reservoirs Supplying Water to Madurai

S. No.	Service Reservoir	Storage Capacity (in Lakh Litre)
1	Semparuthi Nagar –NZ1	18
2	Muthuramalinga Puram Bykara	5
3	Moolakarai Sump – SZ9 (Muthuramalingapuram)	3
4	Muniyandi Kovil street	4
5	Balaji Nagar – SZ12	5
6	Harvipatti – SZ10	10
7	Kurinci Nagar – SZ11(Thirunagar)	5
8	Muthupatti Nagar near PS	20
9	TVS Park Sathya sai Nagar	25
10	TVS Park Sathya sai Nagar	25
11	Villapuram – SZ6 (Ward – 61)	10
12	MMC Colony –SZ 5	15
13	MMC Colony –SZ4	10
14	Vellakkal – SZ8	11
15	Vaalaithoppu Corporation Ground	25
16	Chinna Anuppanadi –SZ2	15
17	Gurunathar Koil – S Z3 (Chinnthamani)	12
18	Thideer Nagar	30
19	Sellur Kanmai – Composting yard Sector 6	25
20	Kattabomman Street (Sivagami Street) Sellur Kanmai – Composting yard Sector 6	25
21	Indranagar – NZ2	15
22	Park Town – NZ5	13
23	EB Colony – NZ7	18

S. No.	Service Reservoir	Storage Capacity (in Lakh Litre)
24	Poriyalar Nagar	7
25	Bharath Nagar	20
26	Island Nagar – NZ8	12
27	Sellur Lorry Stand	30
28	SMP Colony near Mayor Bungalow (MPS site)	10
29	Shenpagathottam – NZ 14	25
30	KK Nagar (Central Veg. Market)	12
31	Kochadai back side of Pumping station	25
32	VKP Nagar Madakulam	20
33	Arappalayam Two wheeler Parking	12
34	AVSS Hospital behind MPS Santhaipettai	20
35	Meenakshi Nagar	15
36	Ulaganeri – NZ 11 (Ward – 28)	12
37	Pandian Nagar – NZ12	25
	Total	589

Table A14.3: Quality Standards of Raw Water and Treated Water

S. No.	Parameters	Raw Water Range	Treated water range
1	pH	7.5	7.1
2	TDS	154	159
3	Turbidity	6.0	0.4
4	Total Hardness	96	90
5	Total Alkalinity	90	86
6	Total Chloride	16	19
7	Residual Chlorine	1.6	6.2
8	Sulphates	7	3

9	Iron	Nil	Nil
10	Fluorides	Nil	Nil

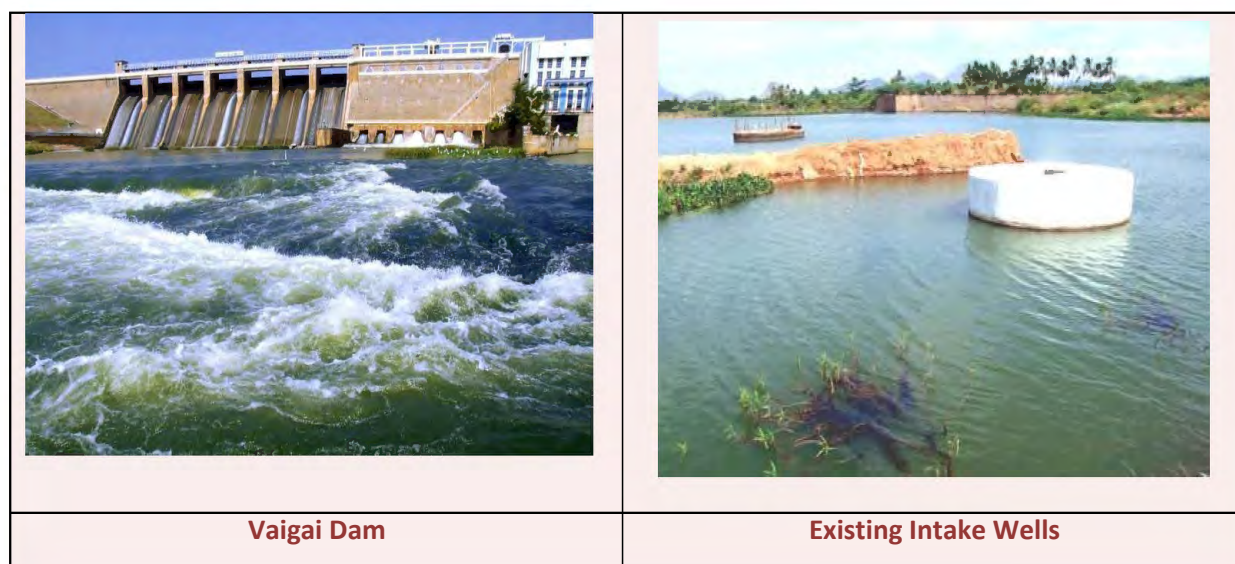
Improvements Made

Under JnNURM funds, three numbers of check dams were constructed in River Vaigai at the following locations:

- (i) Melakkal -To recharge the wells near Melakkal and Thatchampathu.
- (ii) Kochadai - To recharge the wells & Gallery in Kochadai.
- (iii) Thiruppuvanam - To recharge the collector wells in Manalur & Thiruppuvanam.

Vaigai Water Supply Scheme

In this scheme intake well is located about 1.5 km downstream of Vaigai dam on the Vaigai River near Pickup weir. Full yield / supply throughout the year can be obtained through the scheme. This scheme was implemented as Line-I and Line-II in the year 1995 and 2009 respectively. Line-I, having design capacity of 68 MLD, supplies water to the south of River Vaigai (51 Wards) and Line-II with design capacity of 47.00 MLD supplies water to North of River Vaigai (21 Wards) of the Core City Madurai Municipal Corporation. The Corporation is getting about 115 MLD water from the scheme. Schematic of this scheme is shown below.





Madurai Municipal Corporation is mainly depending on Vaigai Dam surface water as source for its drinking water supply needs. As per the Government Order 1500 MCFT /year of water from Vaigai Dam is allotted to Madurai Municipal Corporation for Drinking water supply purpose. The complete quantity of 1500 MCFT / year is now drawn from Vaigai Dam through pipe lines.

Transmission Mains

Raw water from scheme Line - I & Line - II for Vaigai WSS from Vaigai Dam are conveyed to the WTP at Pannaipatti by gravity, through 1100mm PSC transmission main for a length of 16.60 km. Similar to the existing scheme, for the proposed third water supply scheme, raw water from Vaigai Dam is conveyed to the proposed WTP at Pannaipatti through 1067 mm X 8.8 mm thick mild steel pipes for a length of 16.85 km.

In the second water supply scheme, for conveying 47 MLD 1100 mm PSC pipes were designed and executed similar to the first water supply scheme of capacity 71.6 MLD with the hope of getting additional quantity of water from Vaigai Dam. In the above pipe line some leaks were observed in the alignment and the raw water is allowed to bypass the aerator to reduce the number of leaks and to improve the drawal. In the raw water alignment, people who involved in agricultural field, try to tamper the valves and joints when the monsoon fails.

Water Treatment Plant

The existing water treatment plant for Madurai Municipal Corporation was commissioned in the year of 1995 at Pannaipatti village located 16.60 km away from Vaigai reservoir. The capacity of WTP is 118.6MLD (71.60MLD for Line-I & 47.00 MLD for Line-II). The treatment facility is based on the conventional system comprising Aeration, Clariflocculation and Rapid Sand Filtration. The details of Water Treatment Plant are shown below:



Table A14.4: Details of Water Treatment Plant

S. No.	Components	Description
I	Water Treatment Plant	
1	Components	Cascade Aerator – 2 No (10 m Dia Collection tray)
		Stilling Basin -2 (6x4x3.3m)
		Dividing Well – 6m Dia
		Flash Mixer – 4 No's – 3.8 m Dia, 4.05m depth
		Clariflocculator -2 no's (with Tube Settlers)
		Rapid Sand Filter Beds -12 No's (4mx6m) (Dual Media Filters)
		Clear water sump -30 LL and 35 m Dia
2	Capacity of WTP	118.6 MLD
II	Clear Water Reservoir	
1	Capacity	30 Lakh Litres
2	Diameter of Sump	35 m
3	Depth of Sump	2.4 m

A critical component in the water supply improvement scheme is the treatment of surface water drawn from Vaigai Dam Reservoir through installation of water treatment facility. Madurai Municipal Corporation has been allotted 1,500 MCFT / year of water for drinking purposes. Out of this allotment, 900 MCFT / year is being drawn through raw water gravity main and treated at a full scale of 71.60 MLD Water Treatment Plant at Pannaipatti installed as part of a World Bank project in the year 1995.

For the treatment of 600 MCFT/year (47MLD) of water, the already functioning Treatment plant of capacity 71.6 MLD was upgraded to 118.6 MLD by introducing tube settlers in the existing Clariflocculators and by providing dual media with anthracite coal as filter media in addition to the sand in the existing filter beds. The treatment plant is now capable of handling 118.6 MLD of raw water

Clear Water Transmission Main

The clear water from Pannaipatti WTP is conveyed to Arasaradi pumping station and also for few service reservoirs by gravity and then water is distributed to different service reservoirs from the pumping station. Clear water is transmitted to main pumping station (Booster pumping station) by gravity, through 48.80km along 1100mm Φ and 1000mm Φ PSC mains under Line – I and Line – II respectively. details of clear water transmission main for Madurai Municipal Corporation laid under Vaigai scheme (Line – I and Line – II) is given.

Table A14.5: Details of Clear Water Transmission Mains

S. No.	Components	Description Line-I	Description Line-II
1	Type of main	Gravity main	Gravity main
2	From	Pannaipatti WTP	Pannaipatti WTP
3	To	Madurai Municipal Corporation (Arasaradi)	Madurai Municipal Corporation
4	Dia (mm)	1100	1000/900/800
5	Length (km)	48.8	48.8
6	Capacity (MLD)	68	47
7	Material	PSC	PSC/GRP
8	Break pressure tank (lakh litres)	5.0	5.0
9	Year of laying	1995	2009

Pumping Machinery

The details of pumping machinery from source to distribution are presented in the following Tables.

Table A14.6: Details of Pumping Machinery at Source

S.No	Location of Pumps	YOC	Pumps/Motors	No's	Age (Years)
1	Kochadai WSS-I	1924	170HP	2	>40
2	Kochadai WSS-I (Collector Well)	1963	135HP	23	>40
3	Melakal WSS	1963	60HP	2	47
4	Manalur and Thirupuvanam WSS	1983	25HP	2	20
		1983	40HP	2	20
		1983	230HP	2	20
5	Thachampattu	1983	7.5HP	6	5

A14.Table 7: Details of Pumping Machinery at Booster Station

S. No.	Location	Particulars	No's	Motor Details	Pump Details	
					Rated Discharge (LPM)	Rated Head(m)
1	Arasaradi Booster Station	South Zone Feeder	1W+1S	120HP	10896	24
2	Arasaradi Booster Station	North Zone Feeder	1W+1S	150HP	9500	49
3	Arasaradi Booster Station	North Zone Feeder (VWS Scheme)	1W+1S	270HP	19000	49

Service Reservoirs

The Water from Vaigai Dam Reservoir is supplied to North and South zones of Madurai City. The northern part of Madurai i.e., the area above the Vaigai River is called North zone and the area below Vaigai River is called South zone. The entire distribution system is divided in to 28 zones. The North zone is divided in to 12 water distribution zones and whereas the South zone consist of 16 distribution zones. The water supply for the city is getting potable water from 28 numbers of water reservoir/OHT's having different capacities. The total storage capacity of the OHT's is 394.5 LL and the details of service reservoirs are given in the Table 8.

Table A14.8: Details of Existing Service Reservoirs

S. No.	Name of the OHT	Capacity (LL)	Year of Const.	MWL (m)	LWL (m)	Avg. GL (m)	Staging Height (m)
North of River Vaigai (21 Wards)							
1	Aruldasapuram	20.00	1992	161.20	157.20	143.20	14
2	Sellur	13.60	1975	158.10	154.10	142.60	12
3	Rajaji Park	18.16	1975	155.50	151.50	139.60	12
4	Anna Nagar – I	2.30	1975	150.43	146.43	134.43	12
5	Anna Nagar – II	6.00	1992	150.43	146.43	134.43	12
6	K.K. Nagar	8.00	1992	155.1	152.10	140.10	12
7	Lotus Tank	1.00	1990	150.8	147.8	140.3	12
8	K. Pudur	15.00	1975	156.02	152.02	140.02	12
9	Race Course	6.80	1975	158.5	154.5	142.5	12
10	Reserve Line	20.00	1992	168.20	164.2	146.20	18
11	Maruthankulam	5.00	2005	159.60	156.10	140.10	16
12	P&T Nagar	10.0	2005	170.12	166.12	150.12	16
	Total Storage	126.00					
South of River Vaigai (51 Wards)							
1	Kochadai	15.00	1992	157.62	153.62	138.62	15.00

S. No.	Name of the OHT	Capacity (LL)	Year of Const.	MWL (m)	LWL (m)	Avg. GL (m)	Staging Height (m)
2	Palanganatham	25.00	1992	153.50	149.40	134.40	15.00
3	Arapalayam	30.00	1992	158.00	154.00	134.68	19.30
4	Sunday Market I	10.00	1992	149.17	145.67	131.67	14.00
5	Sunday Market II	10.00	1992	149.17	145.67	131.67	14.00
6	Tamil Sangam Road	20.00	1992	151.67	147.67	132.67	15.00
7	East Marret Street	20.00	1992	150.00	146.00	131.37	14.60
8	Traveler's Bungalow	30.00	1992	154.00	150.00	131.50	18.50
9	Joseph Park I	15.00	1975	153.50	149.50	129.50	20.00
10	Joseph Park II	15.00	1992	153.00	149.00	129.50	19.50
11	New Ramnad Road	15.00	1992	147.90	143.90	129.90	14.00
12	Jhansi Rani Park	20.00	1975	156.00	152.00	137.00	15.00
13	Arasaradi	11.50	1975	150.40	146.40	136.35	10.05
14	H.M.S.Colony	16.00	2005	165.19	161.19	145.19	16
15	Sundararajapuram	16.00	2005	160.05	156.05	140.05	16
16	Keeraithurai	16.00	2005	157.01	153.01	137.01	16
	Total Storage	284.50					

Water Distribution System

The potable water supply network covers about 59% of the households in the Madurai Municipal Corporation. The existing water supply consists of about 764km long pressurized pipe network. The distribution system in Madurai Municipal Corporation consist of CI, AC, PVC and PSC pipes with diameter varies from 63mm to 150mm and 1/2inch house service connections are provided to all legal connections. The condition of the pipes in certain areas (core area of the city) is reported to be poor and the carrying capacity reduced due to deposits (incrustation) especially in CI pipes. This may lead to incrustation, friction losses and loss of pressure at the tail end of the distribution network.

House Service Connections: The details of the house service connection are given in the Table 9.

Table A14.9: Details of House Service Connection

S. No.	Type of Connection	No. of Connection
1	Domestic	94,487
2	Non-Domestic	600
3	Commercial	400
	Total	95,487

O&M of the Existing Distribution System

The Madurai Municipal Corporation does the O&M of Water distribution system. Regular records for the valve operations, water levels in towers, water quality are maintained and inlet-outlet meter management are done by this department.

One of the major problems reported in the distribution system is un-equitable supply of water to different parts of a distribution zone. It is observed that the distribution system for a length of about 490Km is more than 30 to 40 years old and these pipes are with incrustations resulting friction and reduction in pressure and flow and also they are very susceptible to damage. It is proposed to revamp the existing distribution system as well as new distribution system for the newly encompassed area (the newly encompassed area not having integrated distribution system).

Water Tariff Structure

The monthly water tariff for domestic house service connection, non-domestic connection as well as for commercial connections is charged on a flat rate as follows.

Table 10: Details of Water Tariff

S. No.	Type of Connection	Monthly Charges	Deposit Charges (Rs.)	Connection Charges (Rs.)
1	Domestic	42.85	1000	1500
2	Non-Domestic/ Intuitionial	85	2000	2000
3	Commercial/ Industrial	150	4000	2500

Appendix 15

Overhead Tanks Details
Appendix 15-Details of OHT site changes

OHT No		Location Name	Ward name/no.	Land detail Khasra no	Classification	Ownership department	Tank Capacity	Area available	Area required	North, South, East, West side bounded details
Old	New									
25	1	Semparuthi Nagar –NZ1	8	TS No. 187/15, 187/6	Vacant Site	MMC	18	2881	529	N – Vacant Area E – Residential W – Residential S – Vacant Area
24	2	Muthuramalinga Puram Bykara	100	TS No. 35/1A	Vacant Site	MMC	5	2236	1050	N – EB Sub Station E – Residential W – Vacant Area S – Private Land
51	3	Moolakarai Sump – SZ9 (Muthuramalingapuram)	99	TS No. 44, 45	Vacant Site	MMC	3	680	81	N – Vacant E – Vacant W – Vacant S – Existing Sump
23	4	Muniyandi Kovil street	100, 75	TS No. 482 XXII – 620	Existing Toilet & Vacant Site	MMC	4	1156	196	N – Vacant Area E – Vacant Area W – Temple S – Vacant Area
50	5	Balaji Nagar – SZ12	99	TS No. 120	Vacant Site	MMC	5	2284	196	N – Water Body E – Residential W – Residential S – Residential
48	6	Harvipatti – SZ10	97	TS No. 151	Vacant Site	MMC	10	1455	400	N – Park E – Corporation Office W – Residential S – Residential
49	7	Kurinchi Nagar – SZ11(Thirunagar)	98	TS No. 105, 106	Vacant Site	MMC	5	4080	200	N – Vacant Area E – Vacant Area W – Residential S – Residential
16	8	Muthupatti Nagar near PS	93	Rs No.	Vacant Site	MMC	20	1000	529	N – Existing Pumping

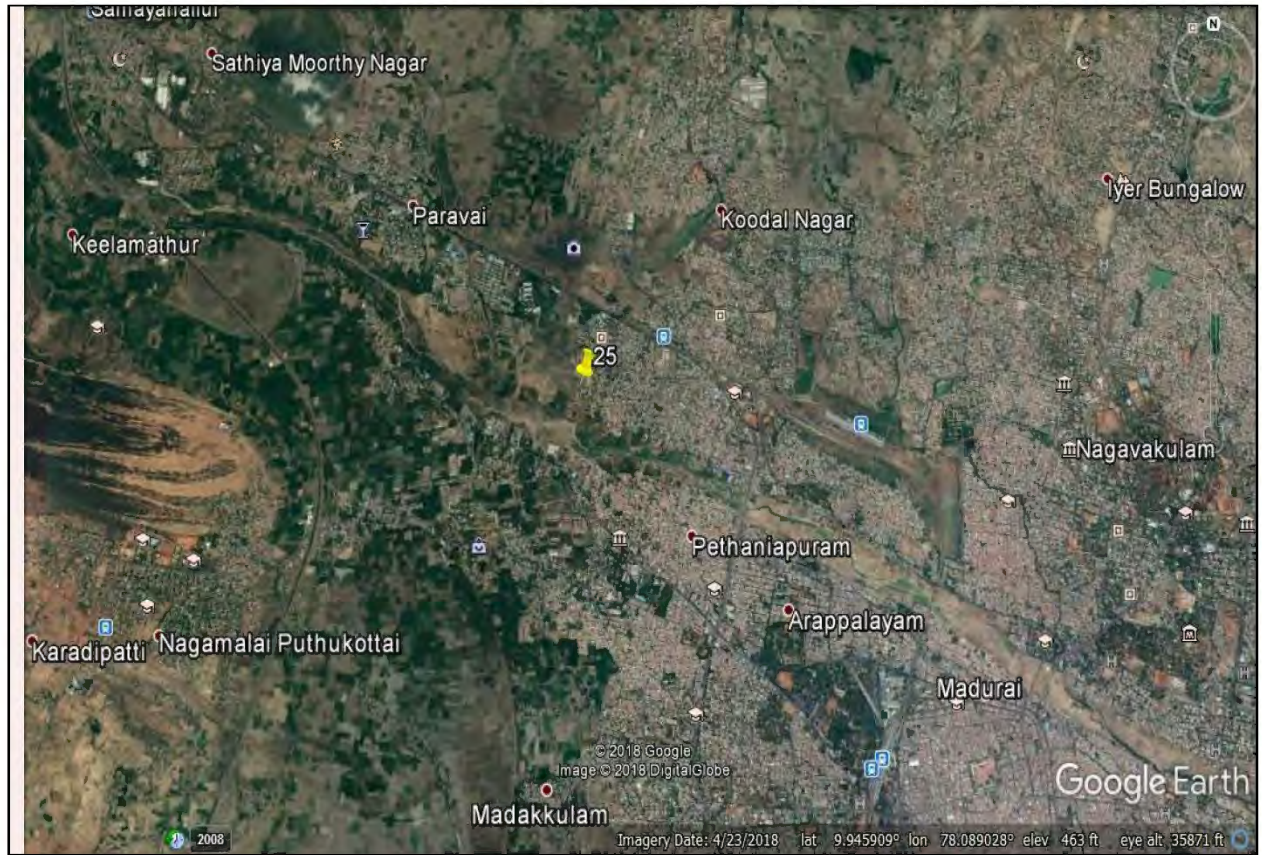
OHT No		Location Name	Ward name/no.	Land detail Khasra no	Classification	Ownership department	Tank Capacity	Area available	Area required	North, South, East, West side bounded details
Old	New									
				440/2, 2						Station E – Commercial Building W – Vacant Land S – Residential
14	9	TVS Park Sathya sai Nagar	77	TS.No. 88, 89	TVS Nagar Park	MMC	25	4771	900	N – Commercial Building E – Commercial Building W – School S – Industry
15	10	TVS Park Sathya sai Nagar	77	TS.No. 88, 89	TVS Nagar Park	MMC	25	4771	735	N – Residential E – Commercial Building W – School S – Industry
45	11	Villapuram – SZ6 (Ward – 61)	61	RS No. 411	Inside of Avaniyapuram WS Pumping Station	MMC	10	3957	400	N – Corporation Building E – Residential W – Industry S – Industry
43	12	MMC Colony –SZ 5	94	TS No. 386	Vacant Site	MMC	15	19360	400	N – Residential E – Vacant Area W – Proposed OHT S – Vacant Area
44	13	MMC Colony –SZ4	94	TS No. 386	Vacant Site	MMC	10	19360	400	N – Residential E – Proposed OHT W – Residential S – Vacant Area
47	14	Vellakkal – SZ8	94	TS No. 156	Vacant Site	MMC	11	7400	528	N – Vacant Area E – Vacant Area W – Vacant Area S – Corp Pump Station

OHT No		Location Name	Ward name/no.	Land detail Khasra no	Classification	Ownership department	Tank Capacity	Area available	Area required	North, South, East, West side bounded details
Old	New									
19	15	Vaalaithoppu Corporation Ground	69	TS No. 2597	Vacant Site	MMC	25	2080	900	N – Residential E – Railway Track W – Commercial Building S – Commercial Building
41	16	Chinna Anuppanadi –SZ2	56	RS No. 198	Vacant Site	MMC	15	1470	400	N – Residential E – Vacant Area W – Vacant Area S – Vacant Area
42	17	Gurunathar Koil – SZ3 (Chinnthamani)	58	TS No. 103/1	Vacant Site	MMC	12	1591	400	N – Urban Health Centre E – Vacant Area W – Temple S – Residential
17	18	Thideer Nagar			Vacant Site	MMC	30			N – Residential E – Residential W – Residential S – Residential
2	19	Sellur Kanmai – Composting yard Sector 6	37	RS 101 to 105	Side of Compost Yard	MMC	25	4089	900	N – Vacant Area E – Vacant Area W – Vacant Area S – Vacant Area
4	20	Sellur Kanmai – Composting yard Sector 6	41	TS No. 3865	Vacant Site	MMC	25	1960	529	N – Residential E – Residential W – Residential S – Residential
29	21	Indranagar – NZ2	23	Ts No. 73/4, 74/2	Side of Vilangudi Kanmoi	MMC	15	2795	400	N – Industry E – Vacant Area W – Vilangudi Kanmoi S – Vilangudi Kanmoi
30	22	Park Town – NZ5	5	TS No. 3865	Meenakshi Nagar Park	MMC	13	1591	400	N – Residential E – Existing OHT

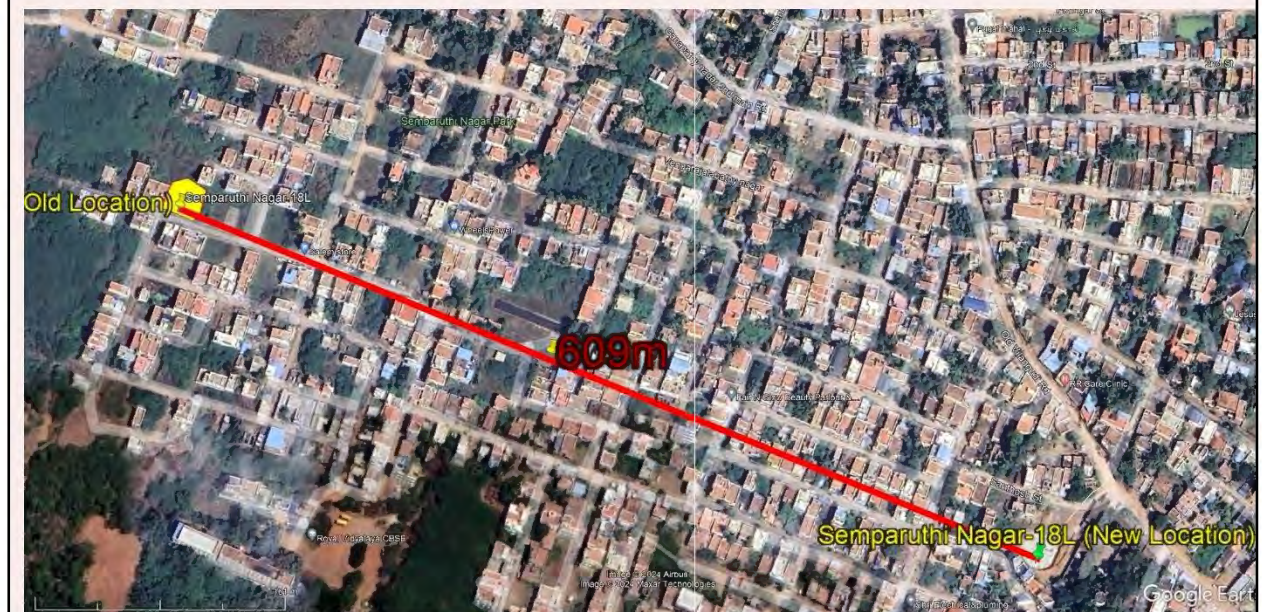
OHT No		Location Name	Ward name/no.	Land detail Khasra no	Classification	Ownership department	Tank Capacity	Area available	Area required	North, South, East, West side bounded details
Old	New									
										W – Residential S – Vacant Area
32	23	EB Colony – NZ7	24	TS No. 79/4	Vacant Site	MMC	18	1088	400	N – Residential E – Vacant Area W – Residential S – Residential
37	24	Poriayar Nagar	25	TS No. 158 (P)/160 (P) B	Vacant Site	MMC	7	1591	774	N – Temple E – Kannanendhal Kanmoi W – Residential S – Residential
34	25	Bharath Nagar	25	TS No. 132/1	Vacant Site	MMC	20	1657	400	N – Residential E – Residential W – Residential S – Vacant Area
36	26	Island Nagar – NZ8	48	TS No. 51/2	Vacant Site	MMC	12	1110	400	N – Residential E – Vacant Area W – Residential & Temple S – Aathikulam Kanmoi
3	27	Sellur Lorry Stand	37	TS No. 1382/2(P)	Corp Lorry Stand	MMC	30	1000	625	N – Lorry Stand E – Institution W – Residential S – Vaigai River
6	28	SMP Colony near Mayor Bungalow (MPS site)	33	TS No. 38/7 to 15	Vacant Site	MMC	10	5026	500	N – Residential E – Exist Sewage Sump W – Residential S – Vaigai River
39	29	Shenpagathottam – NZ 14	33, 34	TS No. 42	Corp Park	MMC	25	3150	550	N – Park E – Residential W – Commercial Building

OHT No		Location Name	Ward name/no.	Land detail Khasra no	Classification	Ownership department	Tank Capacity	Area available	Area required	North, South, East, West side bounded details
Old	New									
										S – Residential
7	30	KK Nagar (Central Veg. Market)	44	TS No. 2236, 2737, 11/10	Vacant Site	MMC	12	3862	225	N – Vacant Area E – Market W – Petrol Bunk S – Residential
13	31	Kochadai back side of Pumping station	22	TS No. 1/1V/15	Vacant Site	MMC	25	4800	576	N – Vaigai River E – Vacant Area W – Residential S – Existing OHT
12	32	VKP Nagar Madakulam	75	TS No. 16/14	Vacant Site	MMC	20	4500	529	N – Vacant Area E – Vacant Area W – Temple S – Vacant Area
10	33	Arappalayam Two wheeler Parking	15	TS No. 970	Vacant Site	MMC	12	1814	900	N – Vaigai River E – Existing OHT W – Vaigai River S – Residential
20	34	AVSS Hospital behind MPS Santhaipettai	52	TS No. 11, V/17	Vacant Site	MMC	20	5284	900	N – Vaigai River E – Residential W – Corporation Land S – Residential
21	35	Meenakshi Nagar	53, 55	TS No. 118	Vacant Site	MMC	15	2860	750	N – Vaigai River E – Service Station W – Residential S – Residential
53	36	Ulaganeri – NZ 11 (Ward – 28)	28	TS No. 21	Vacant Site	MMC	12	2444	900	N – Residential E – Vacant Area W – Residential S – Water Body
54	37	Pandian Nagar – NZ12	29	RS No. 1 (Vandiyur 1 st Pit))	Vacant Site	MMC	25	1184	400	N – Residential E – Residential W – Vandiyur Pond S – Vandiyur Pond

Overhead Tank Site Locations:

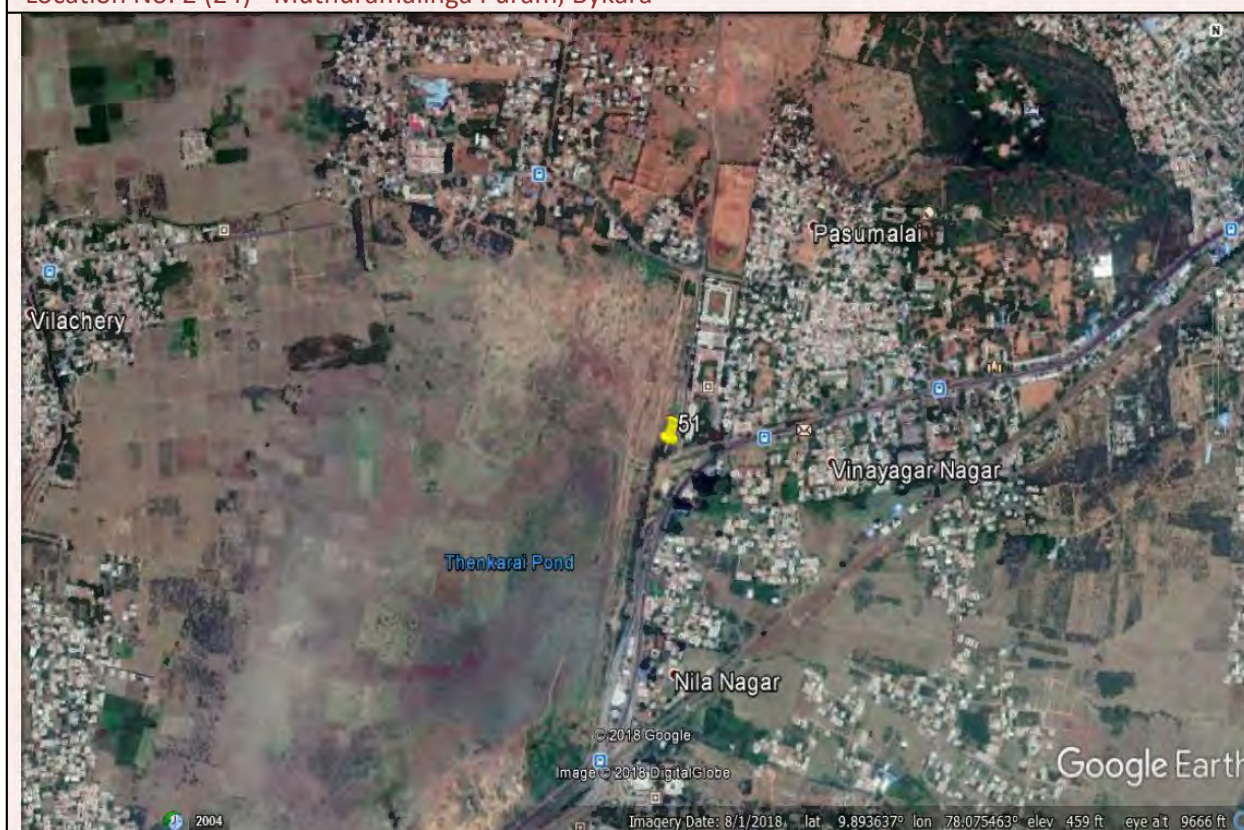


Location No: 1 (25) - Semparuthi Nagar – NZ1

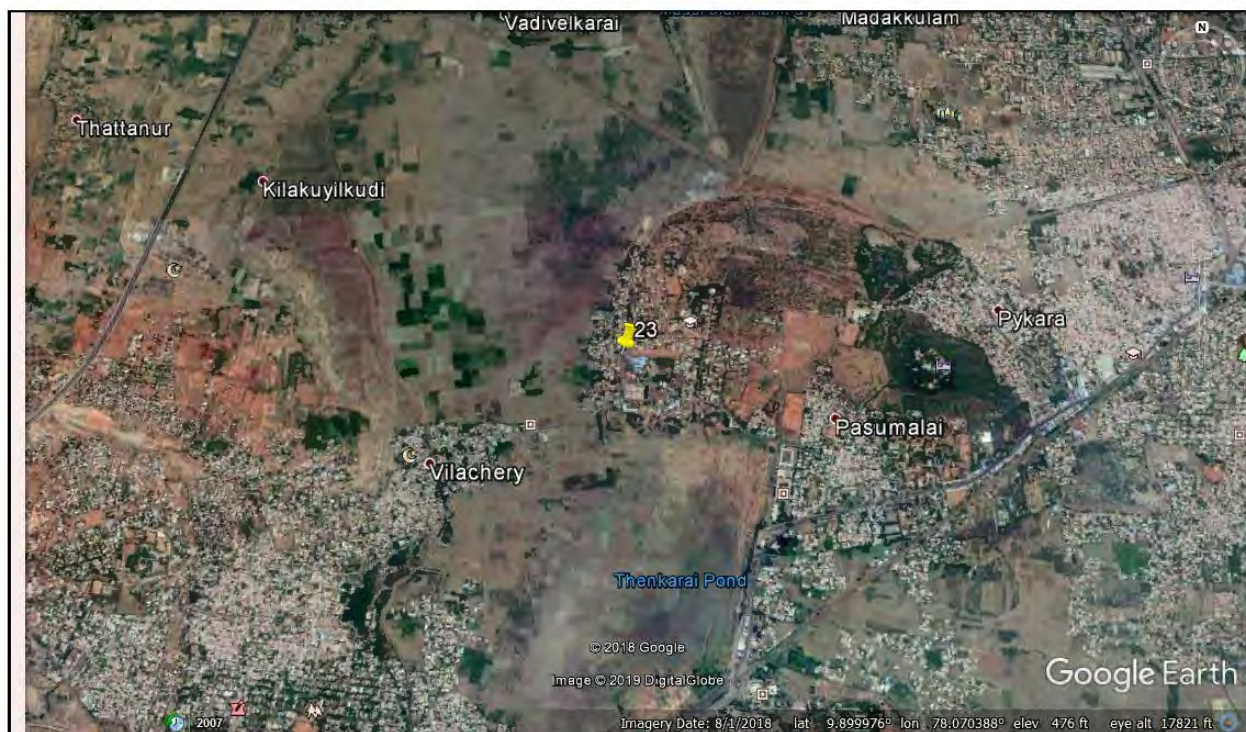




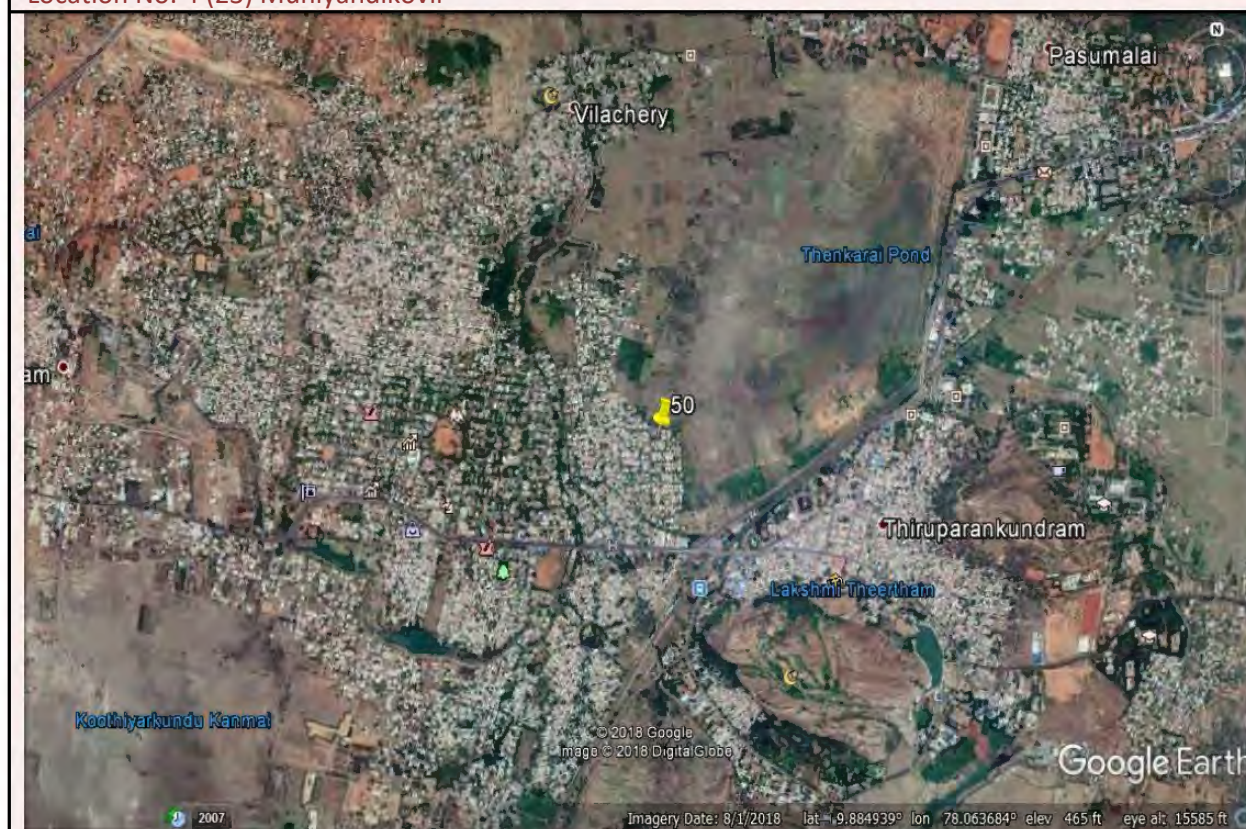
Location No: 2 (24) - Muthuramalinga Puram, Bykara



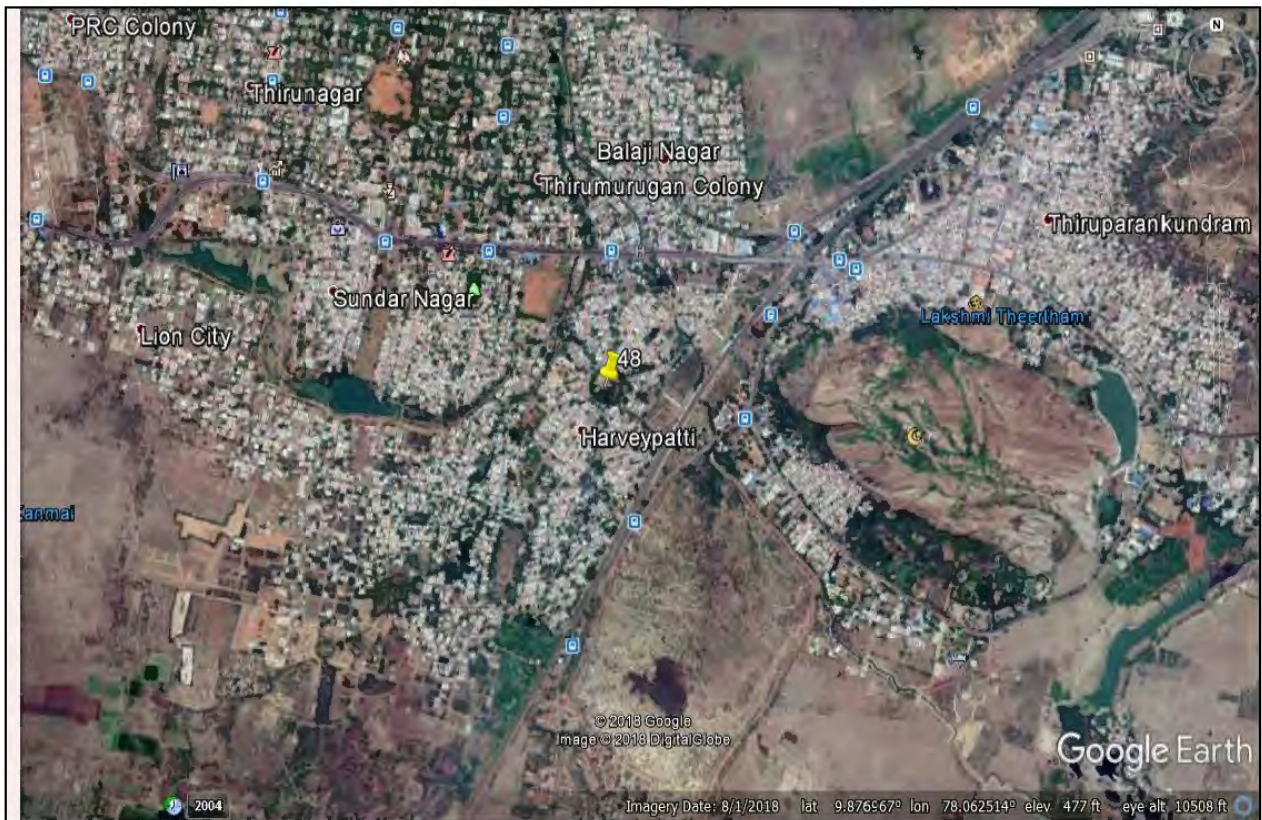
Location No: 3 (51) - Moolakarai Sump - SZ9



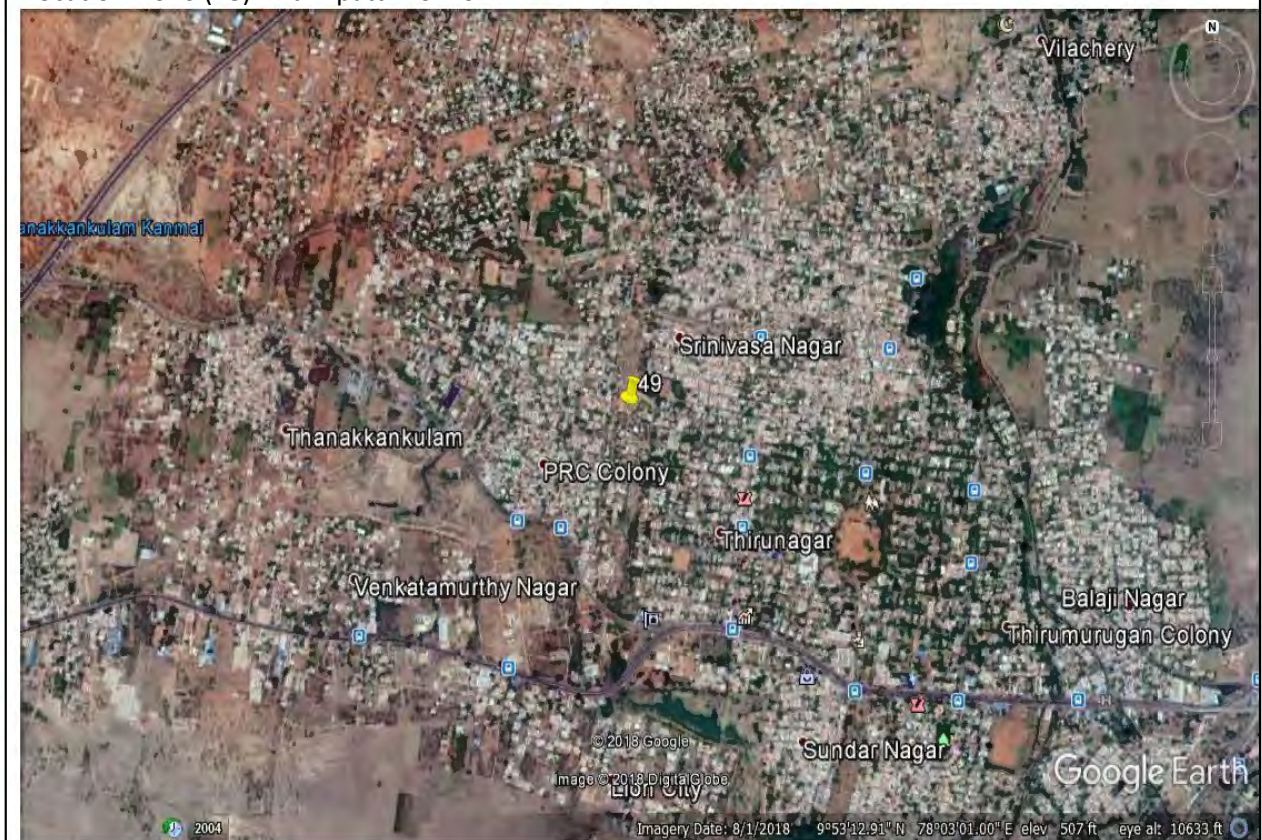
Location No: 4 (23) Muniyandikovil



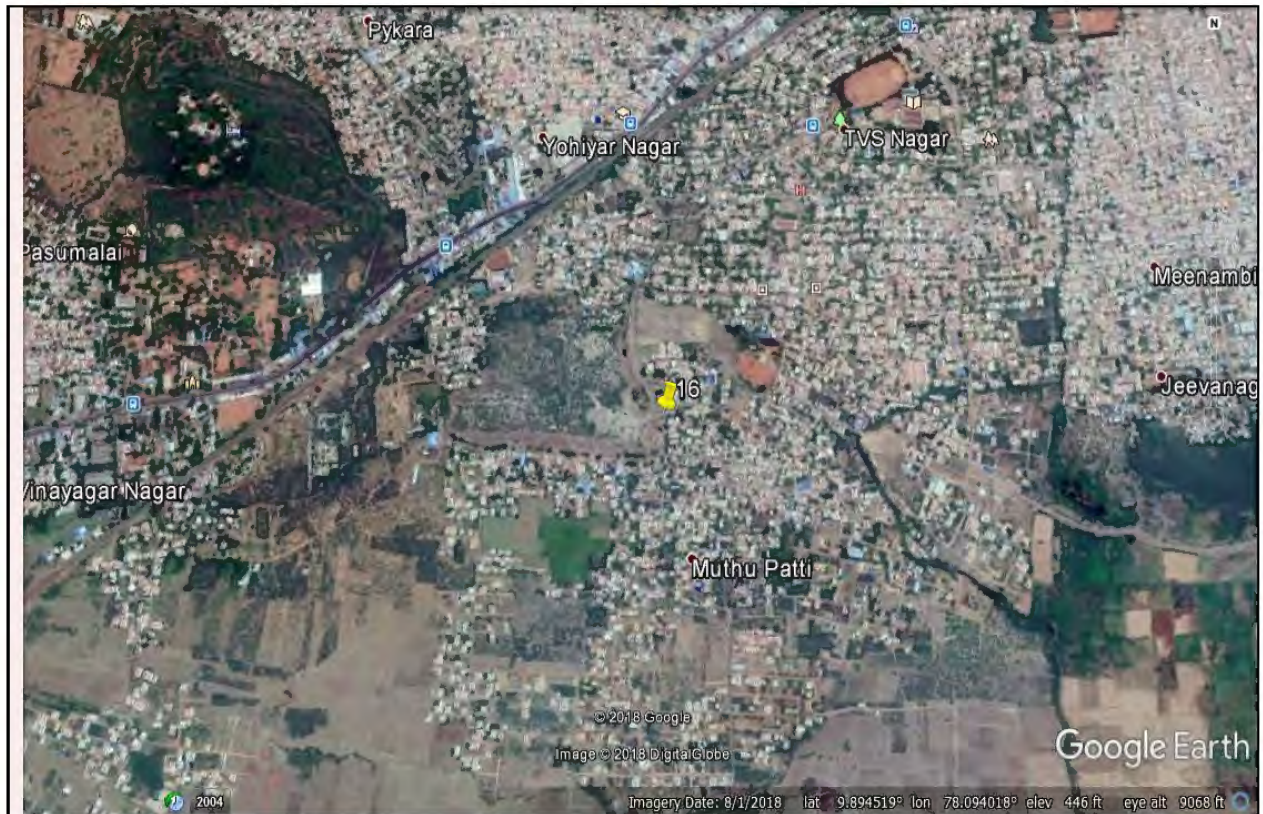
Location No: 5 (50) - Balaji Nagar – SZ12



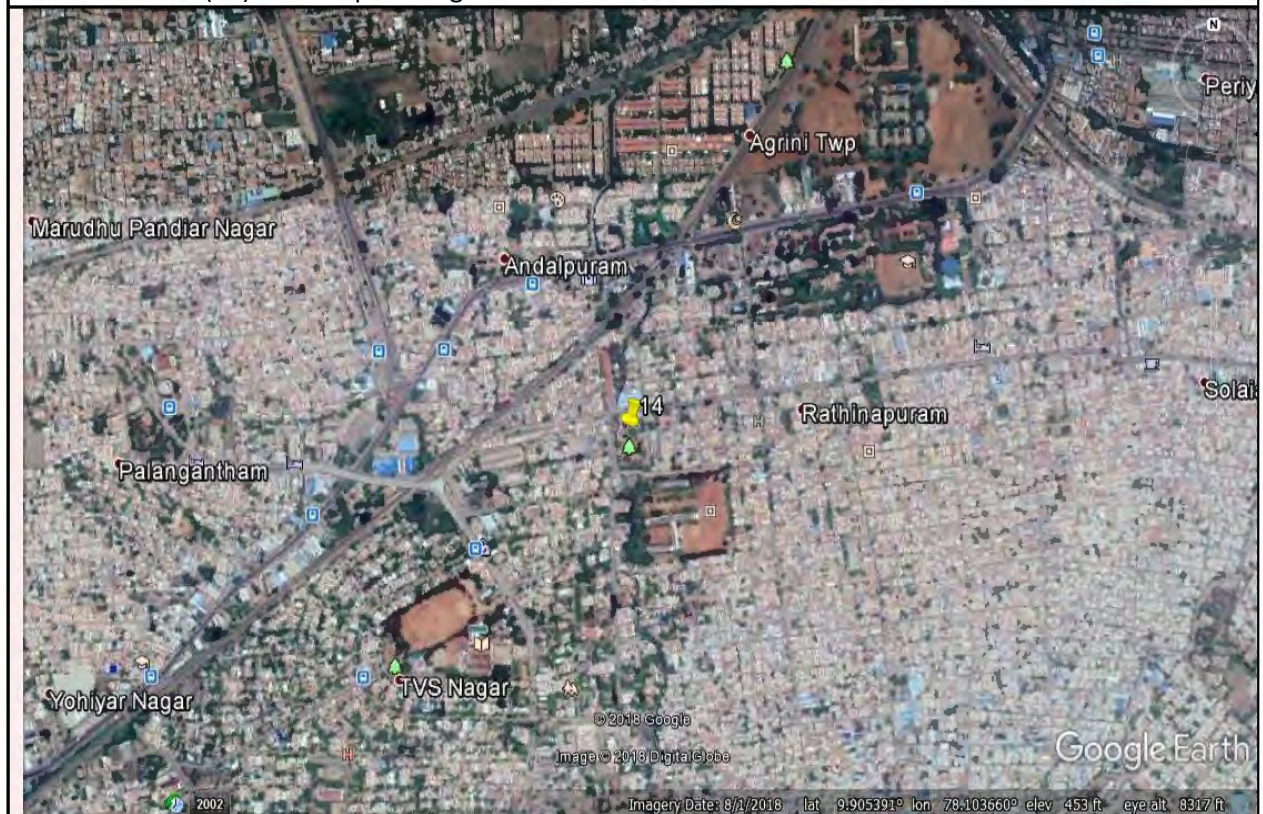
Location No: 6 (48) - Harvipatti – SZ10



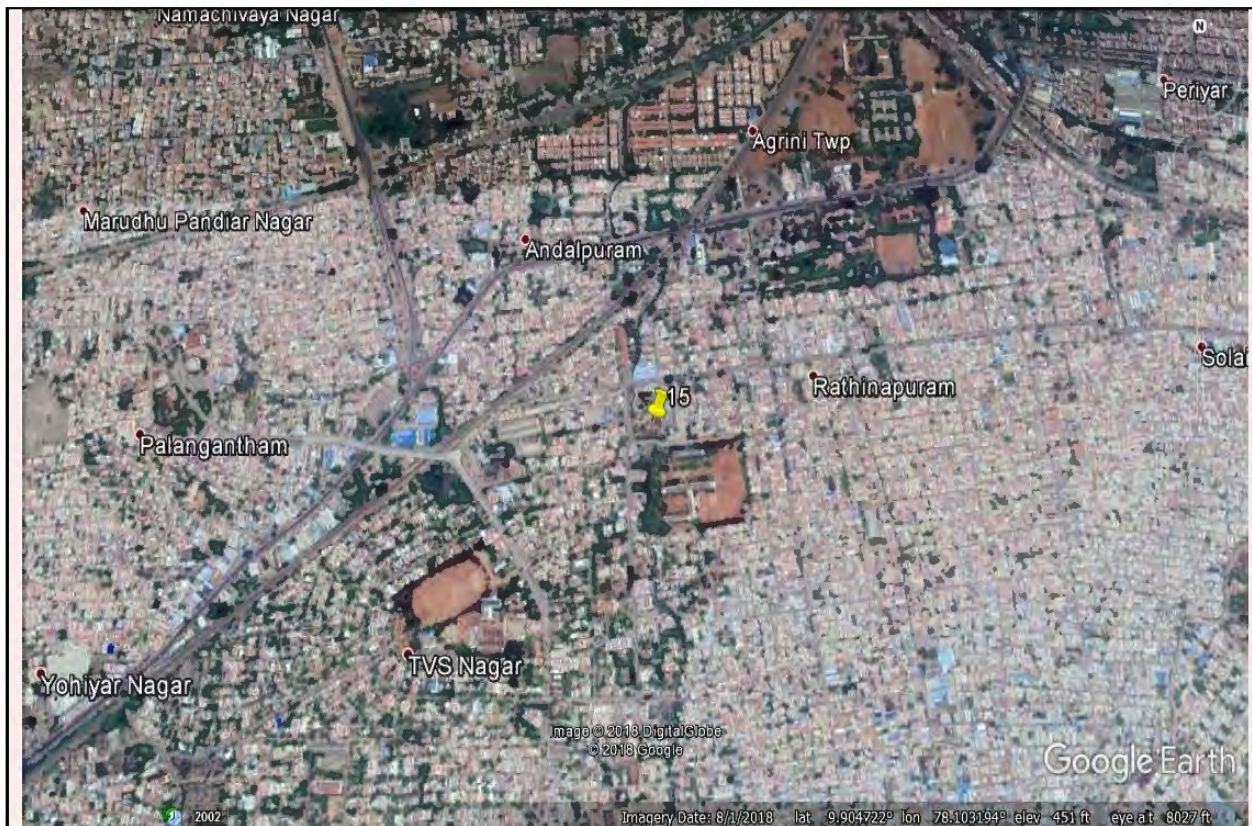
Location No: 7 (49) - Kurinchi Nagar – SZ11 (Thirunagar)



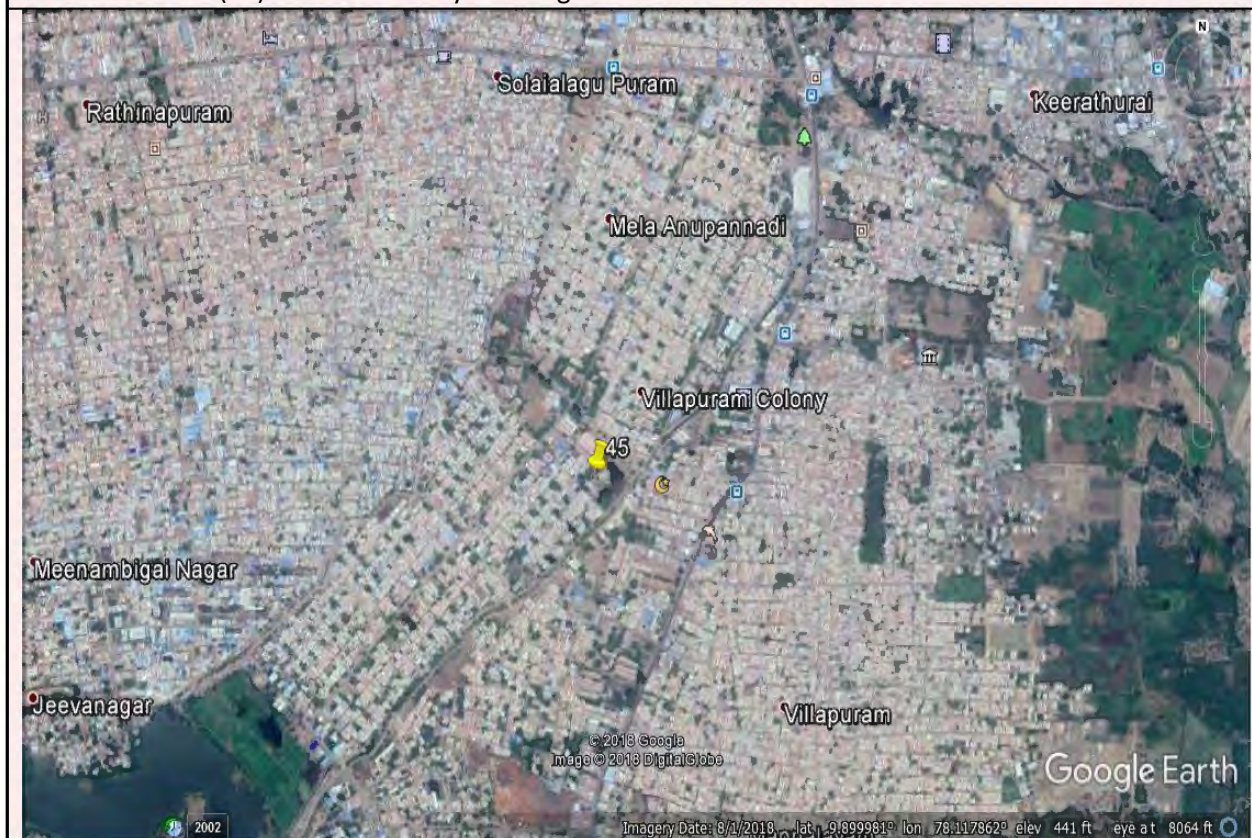
Location No: 8 (16) - Muthupatti Nagar near PS



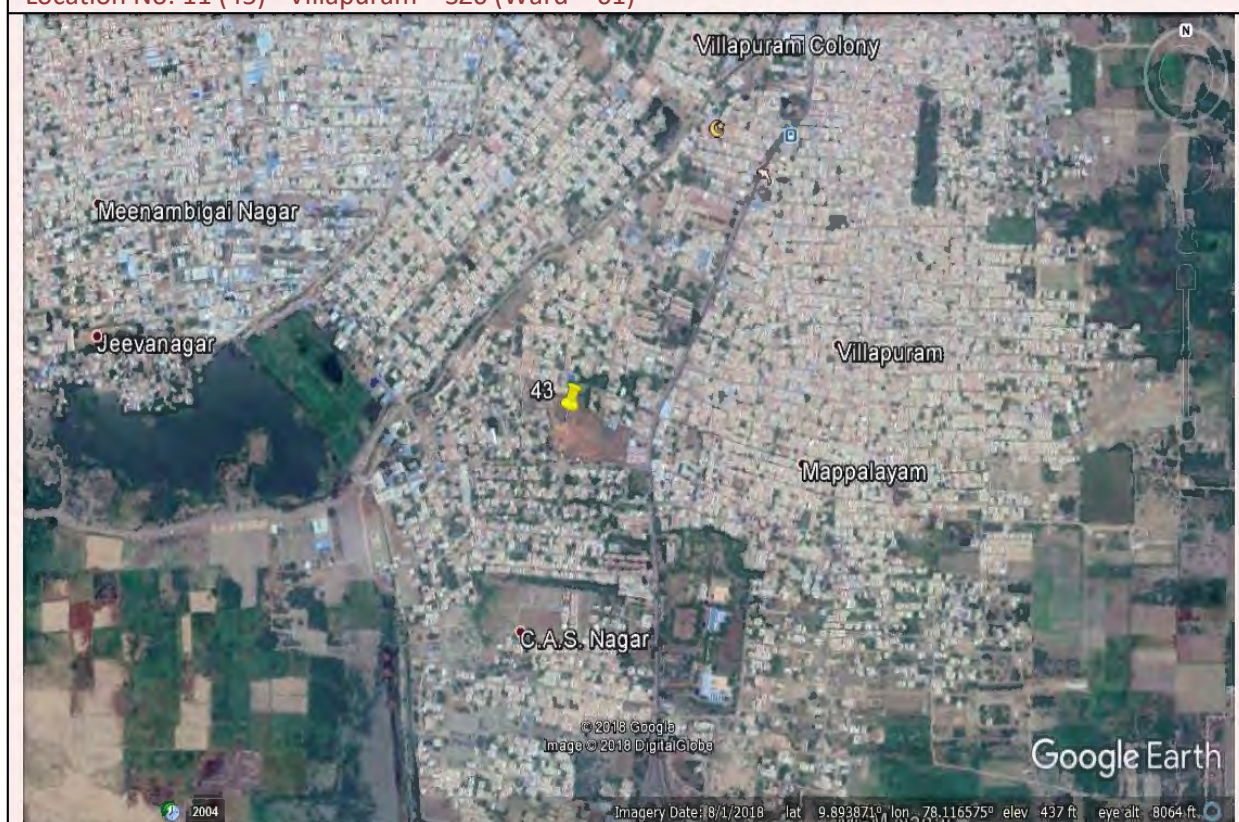
Location No: 9 (14) - TVS Park Sathya sai Nagar



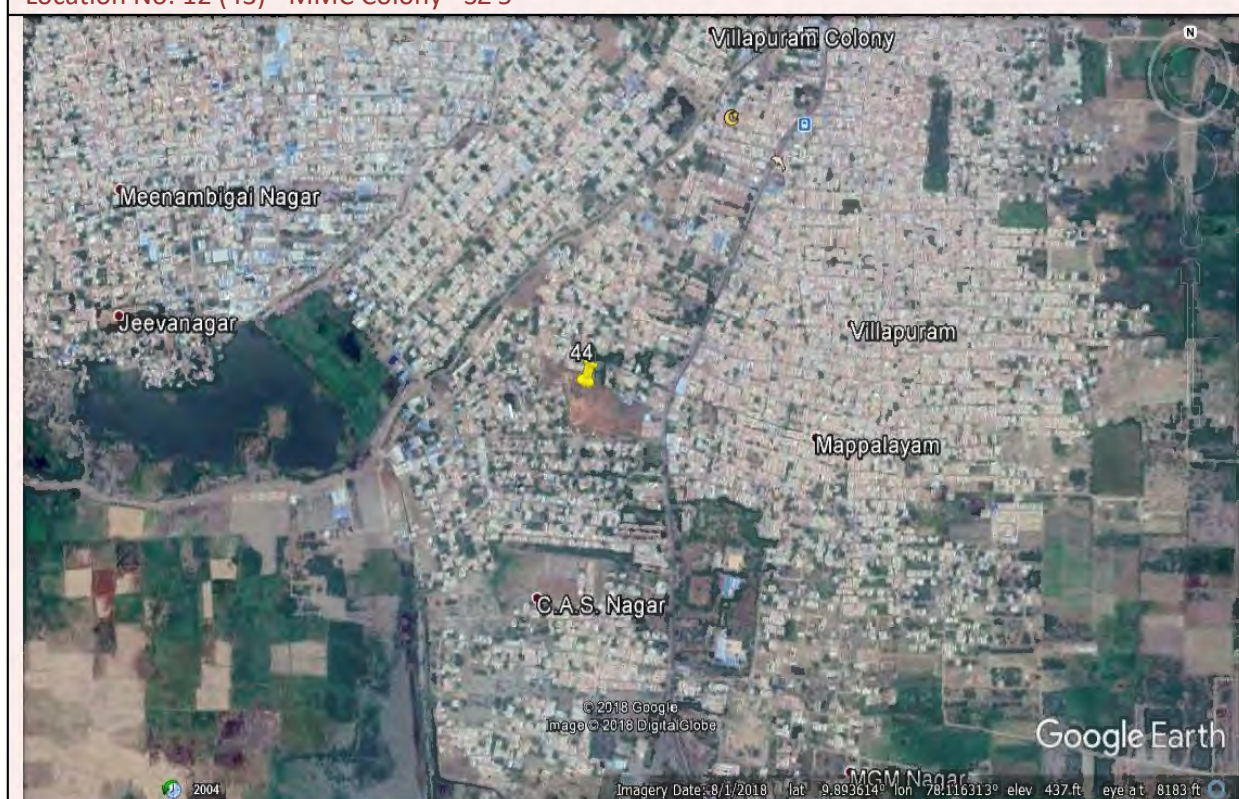
Location No: 10 (15) - TVS Park Sathya sai Nagar



Location No: 11 (45) - Villapuram – SZ6 (Ward – 61)



Location No: 12 (43) - MMC Colony –SZ 5



Location No: 13 (44) - MMC Colony –SZ4



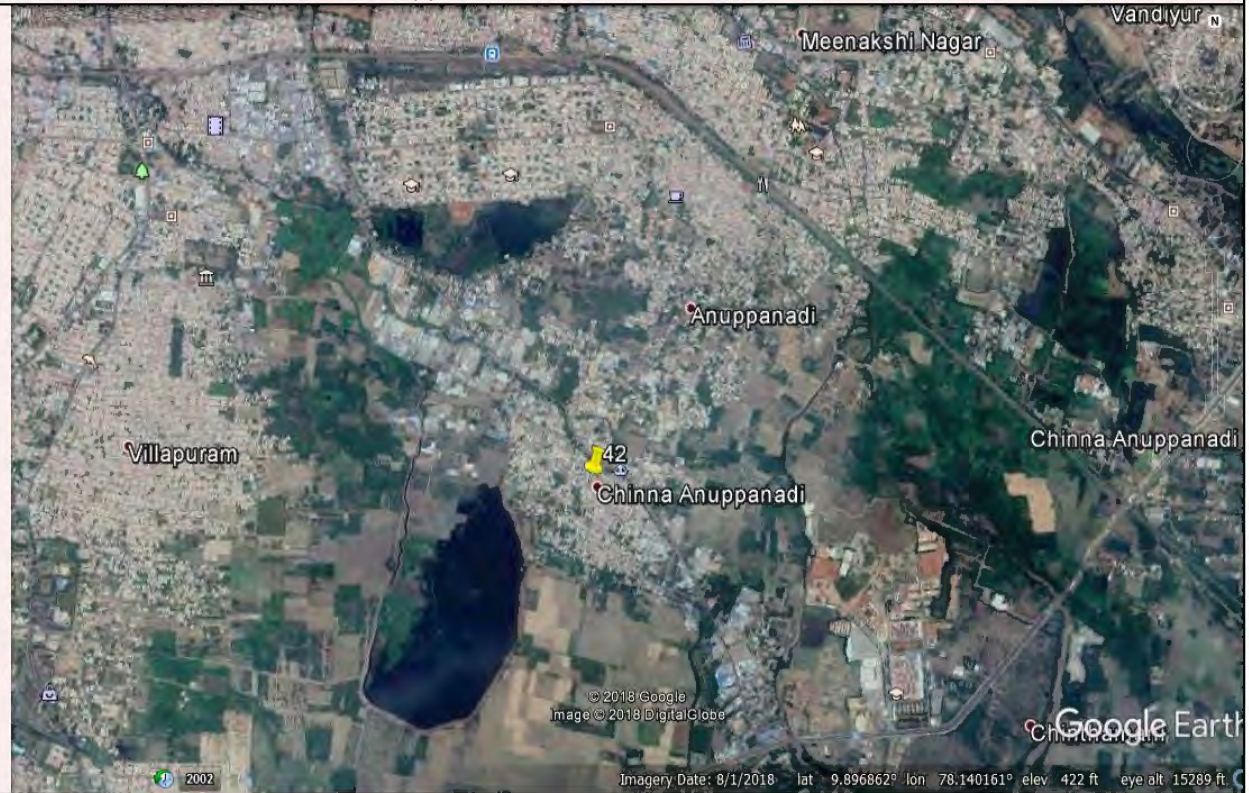
Location No: 14 (47) - Vellakkal



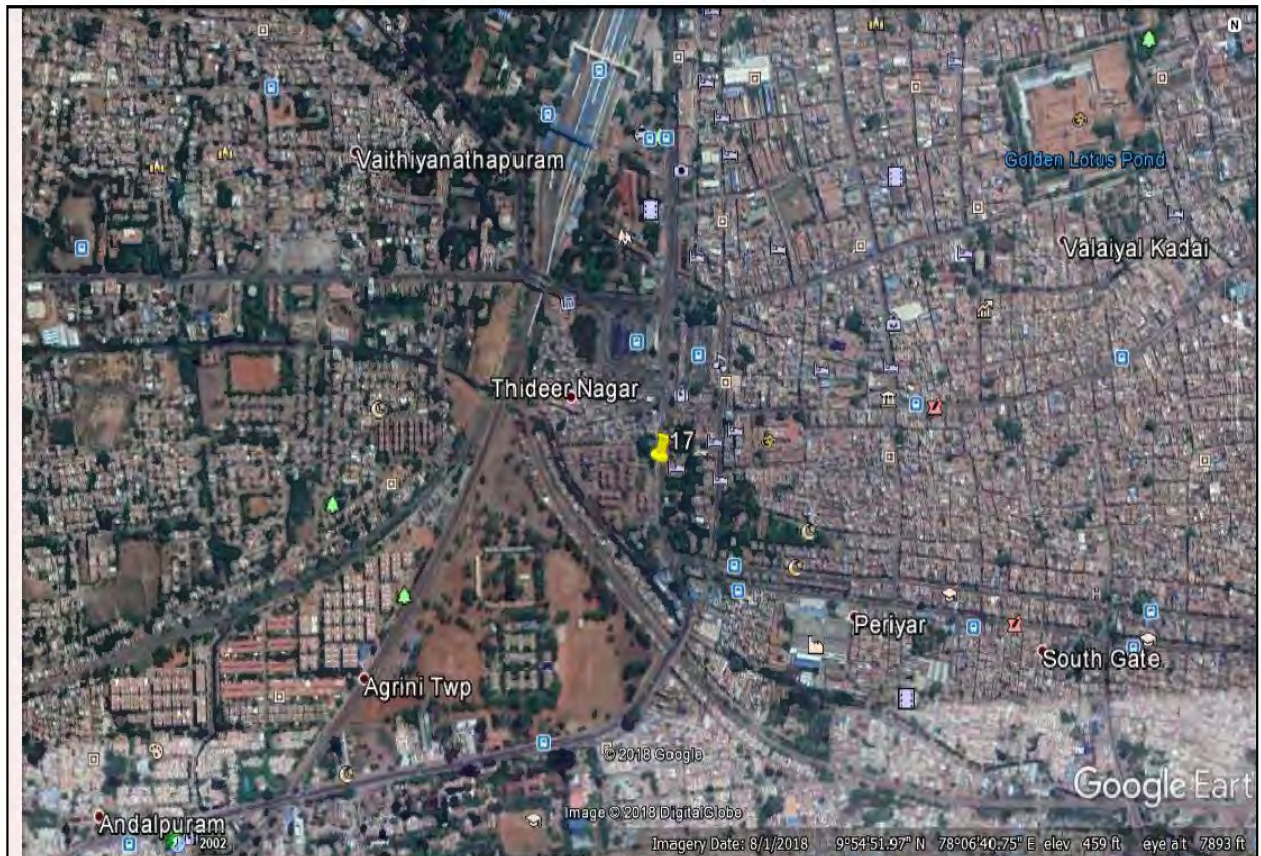
Location No: 15 (19) - Vaalaithoppu Corporation Ground

S

Location No: 16 (41) - Chinna Anuppanadi –SZ2



Location No: 17 (42) - Gurunathar Koil – SZ3 (Chinnthamani)



Location No: 18 (17) - Thideer Nagar



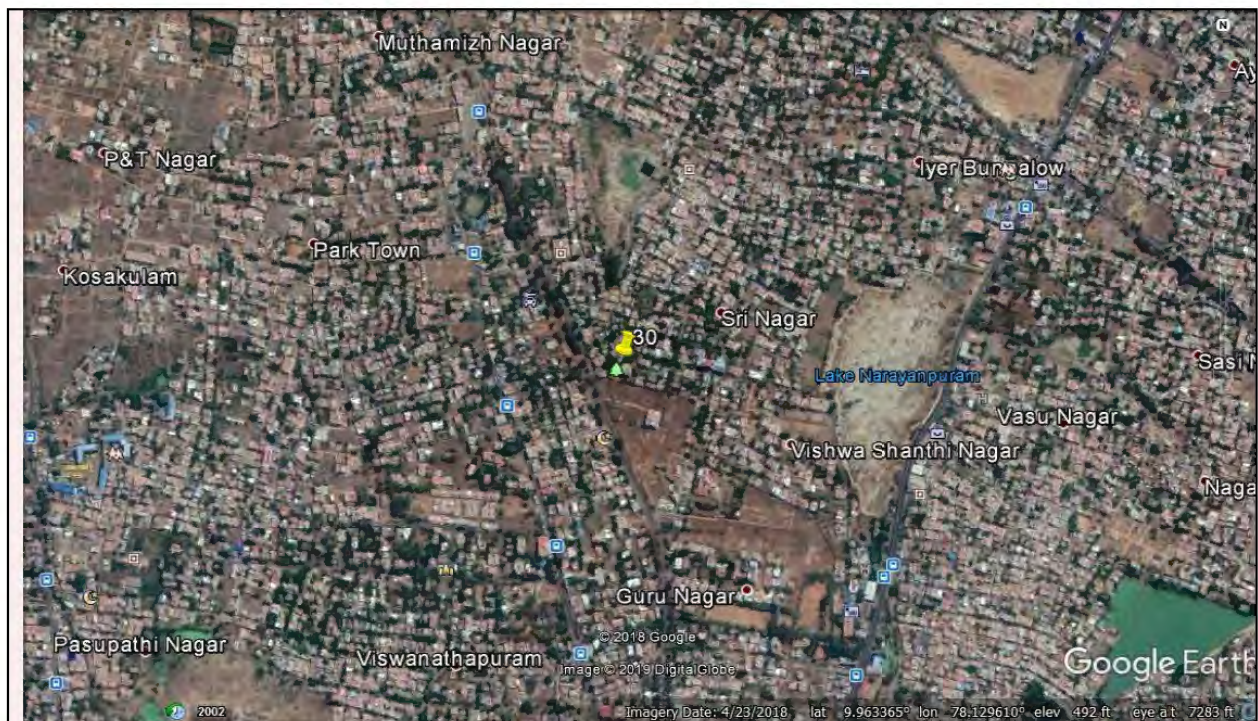
Location No: 19 (2) - Sellur Kanmai



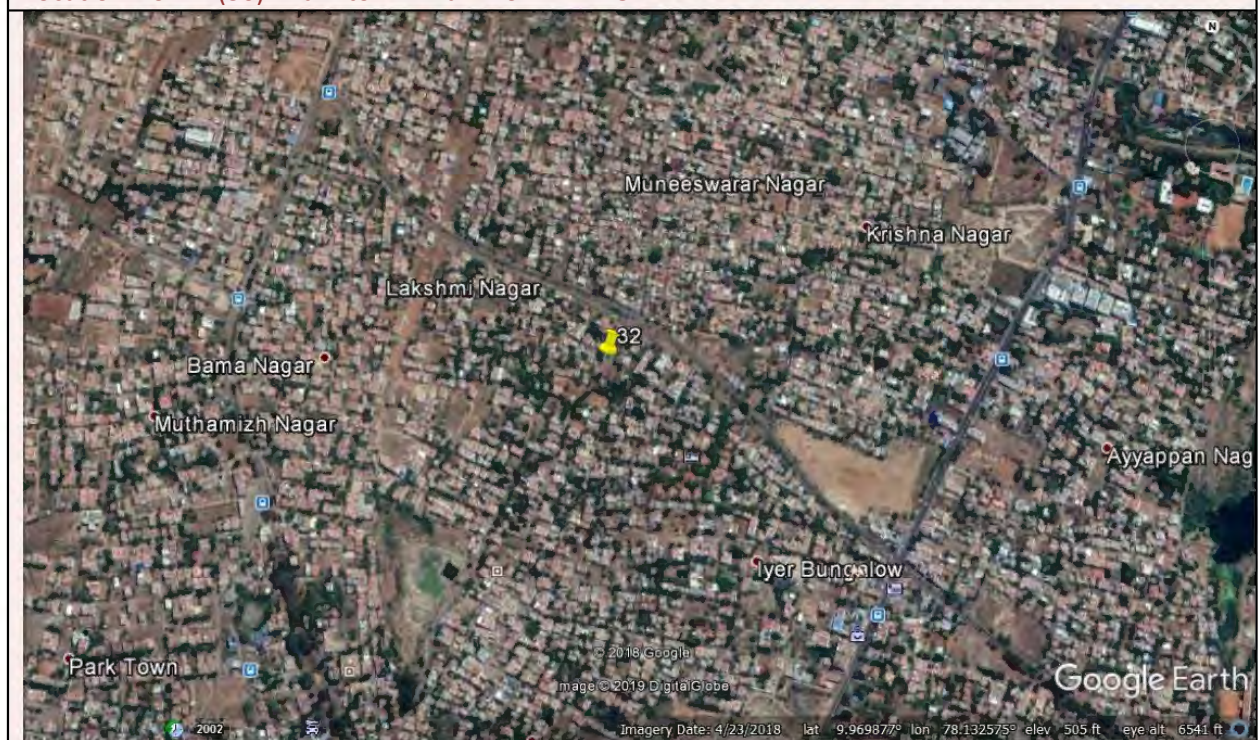
Location No: 20 (4) – Sellur Kanmai



Location No: 21 (29) - Indranagar – NZ2



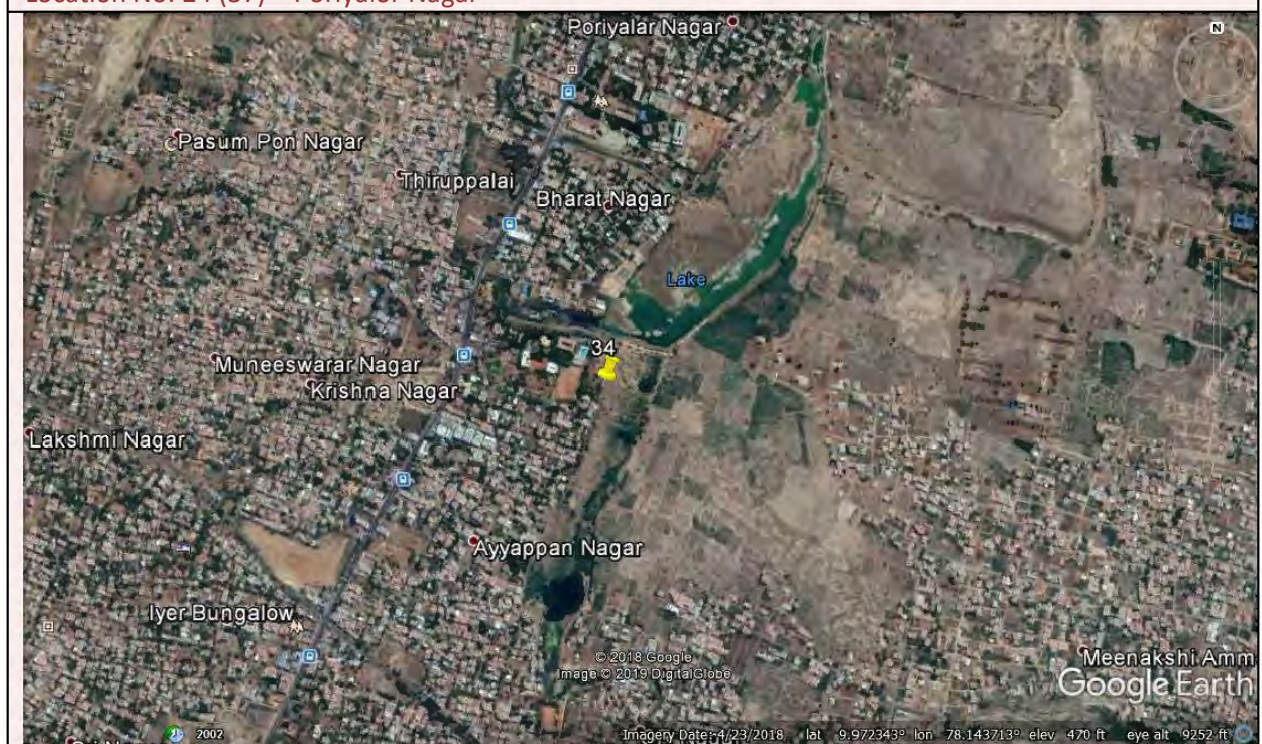
Location No: 22 (30) - Park town - Park Town – NZ5



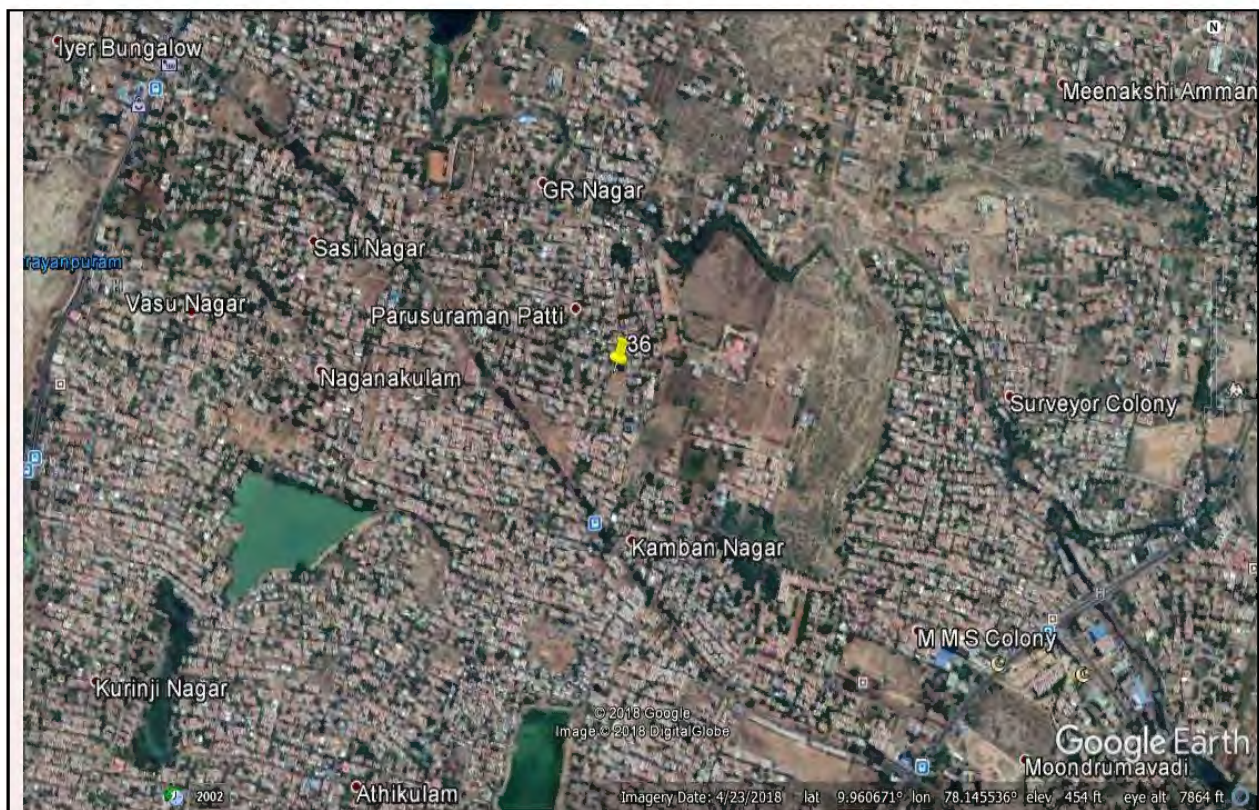
Location No: 23 (32) - EB colony



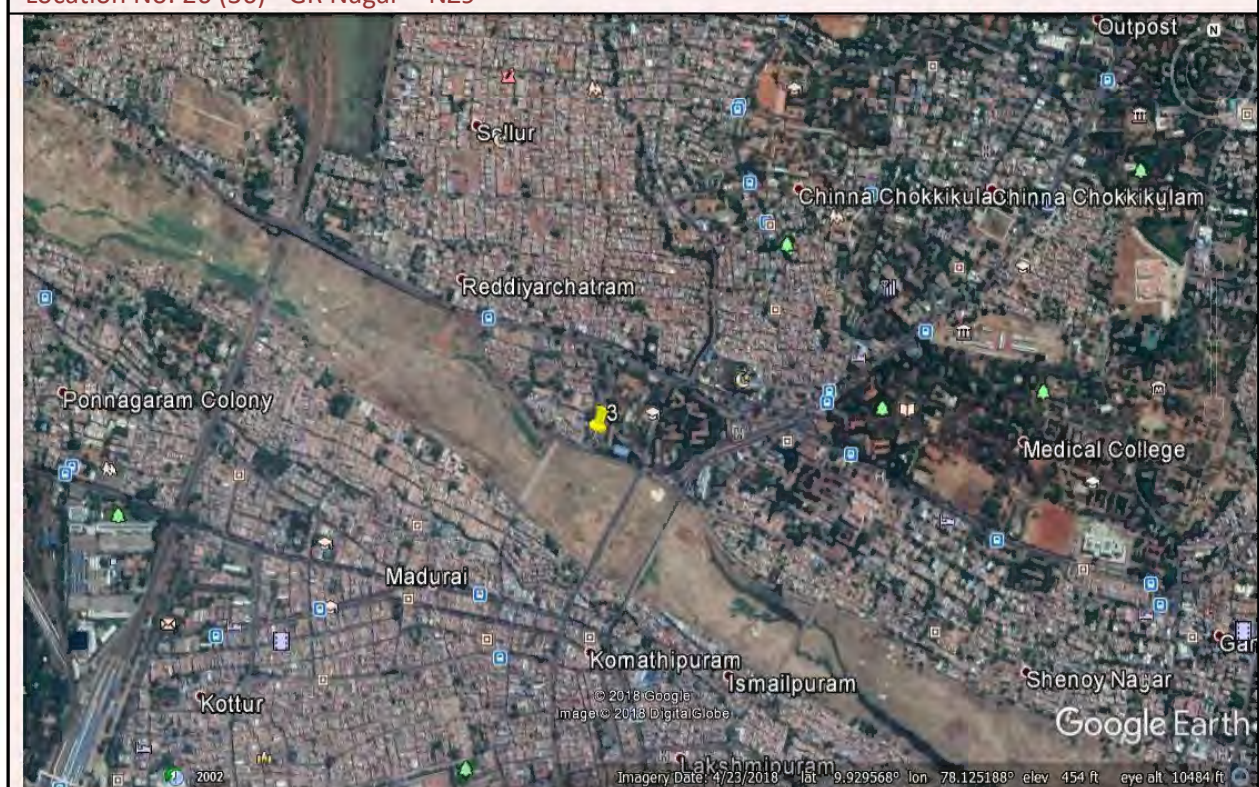
Location No: 24 (37) – Poriyalar Nagar



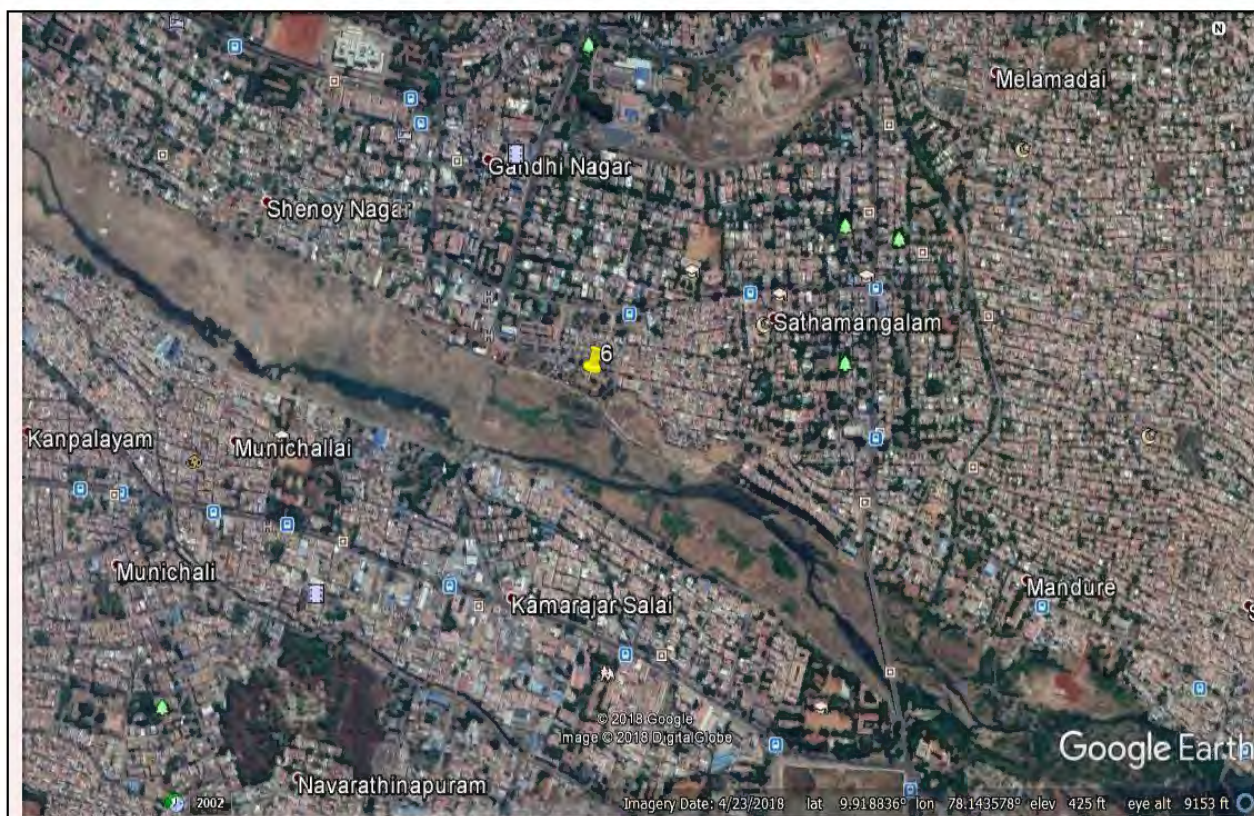
Location No: 25 (34) - Bharath Nagar



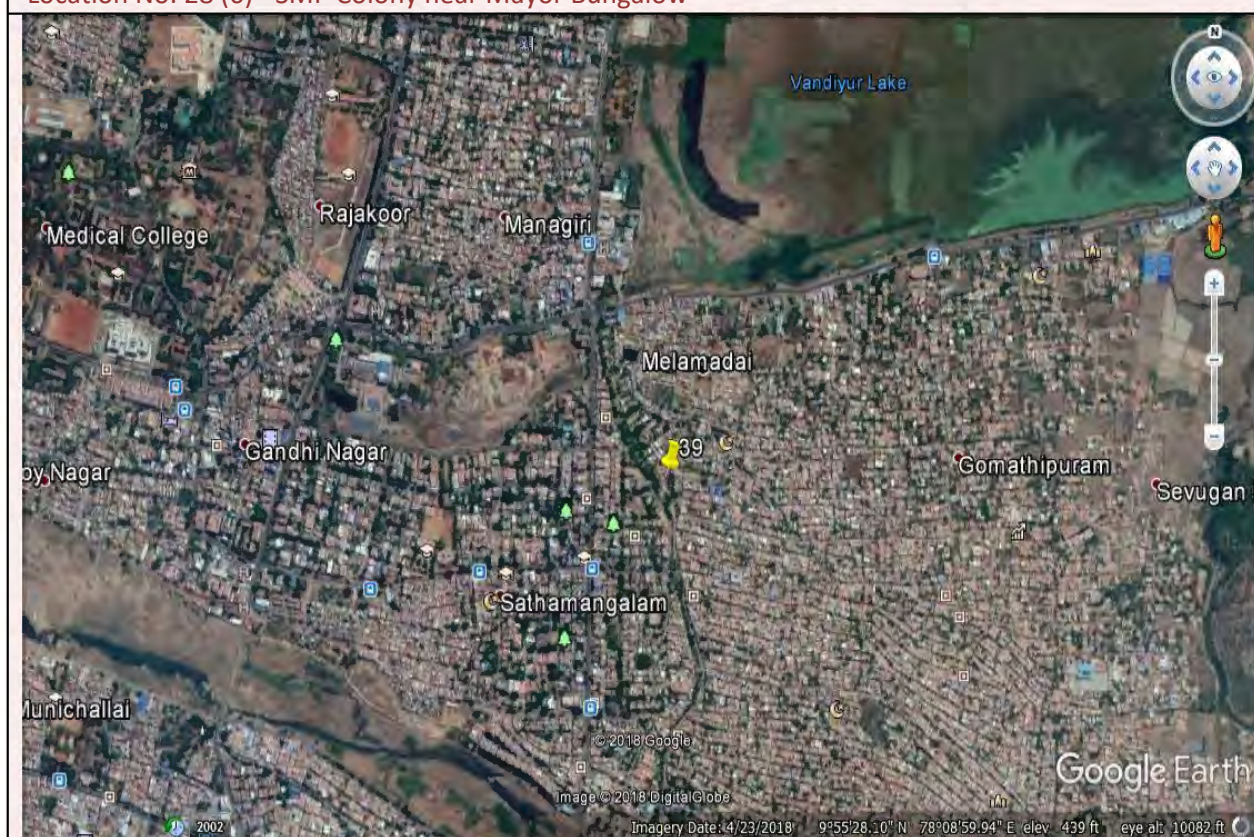
Location No: 26 (36) - GR Nagar – N29



Location No: 27 (3) - Sellur Lorry Stand



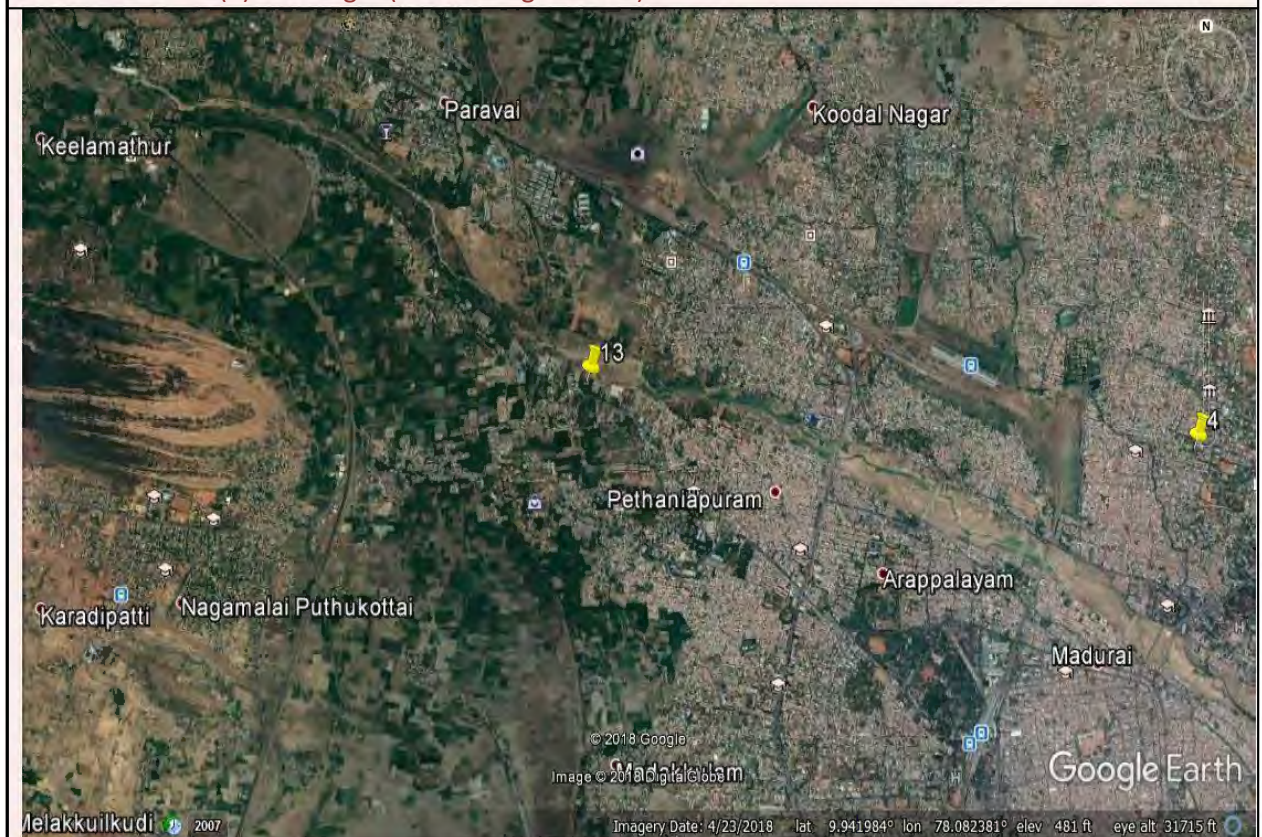
Location No: 28 (6) - SMP Colony near Mayor Bunglow



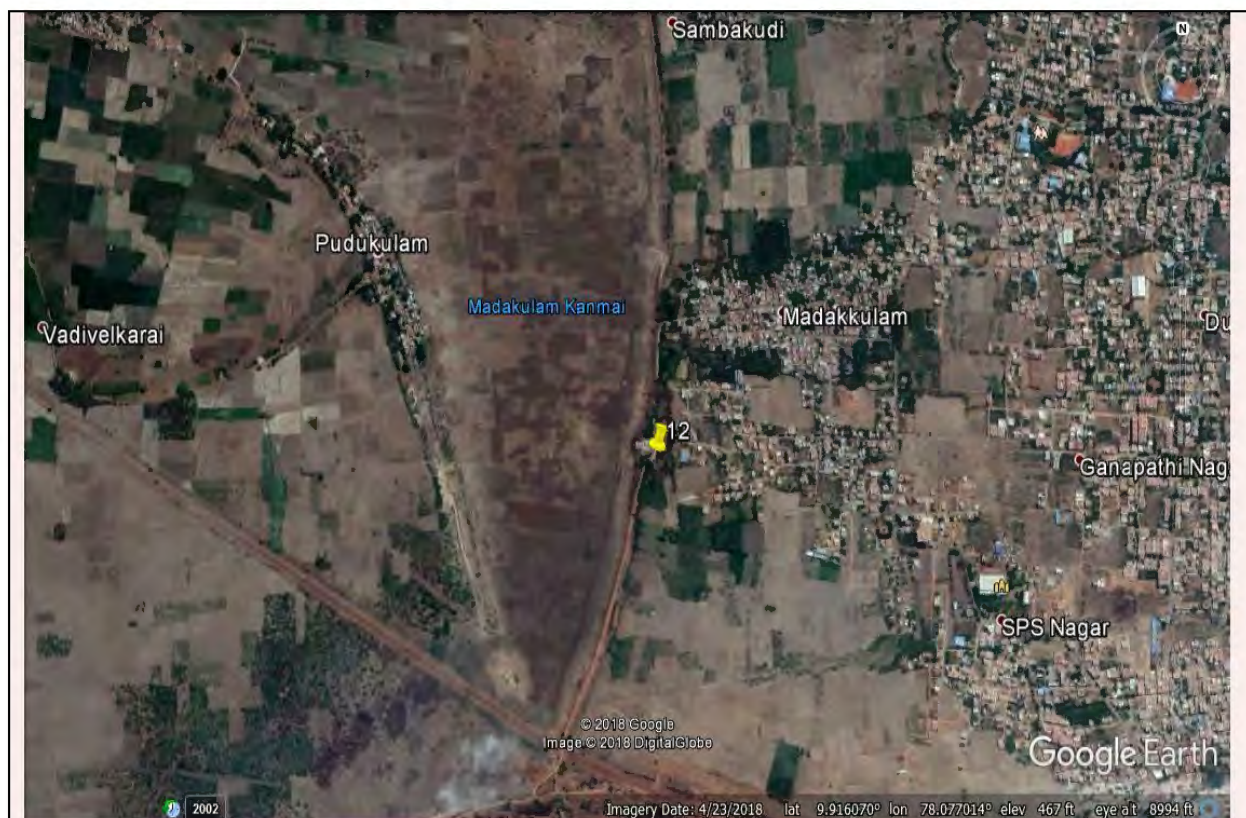
Location No:29 (39) - Shenpagathottam



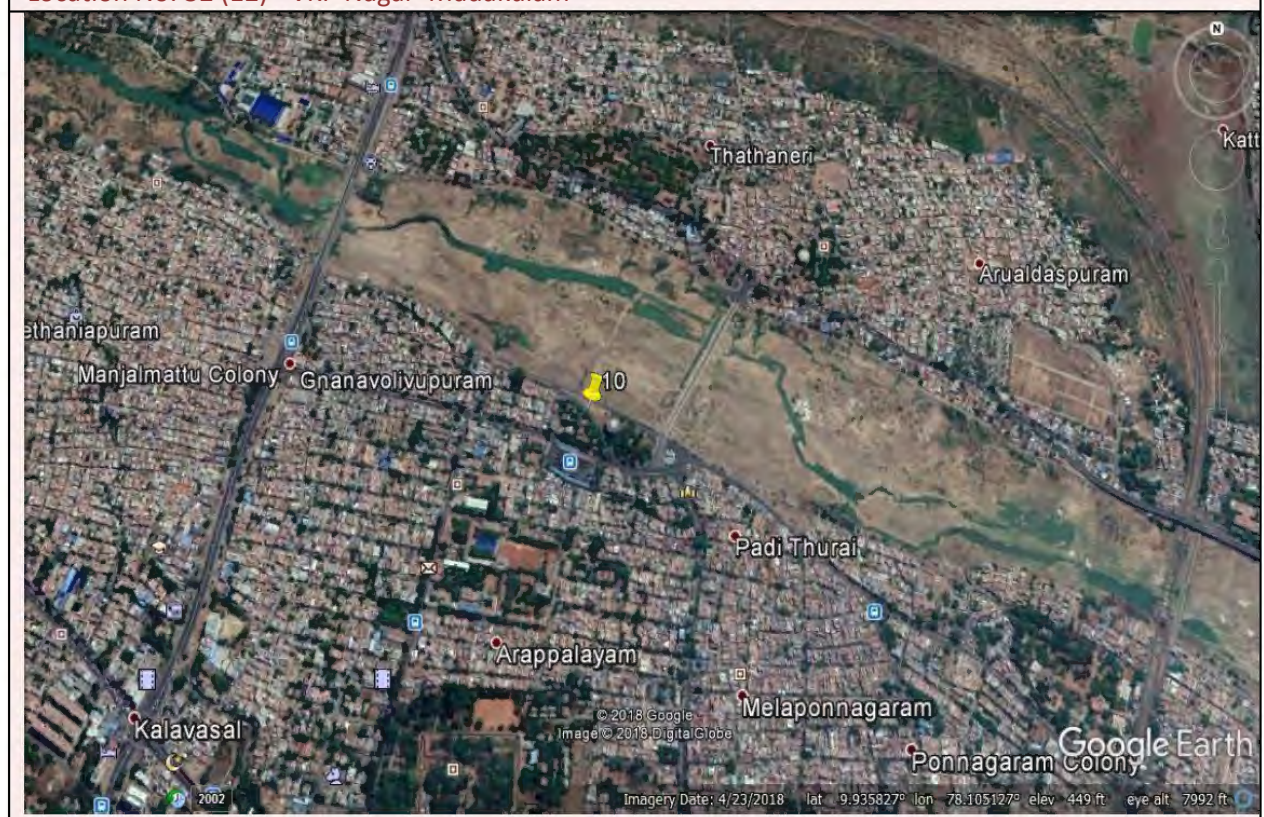
Location No: 30 (7) - KK Nagar (Central Veg. Market)



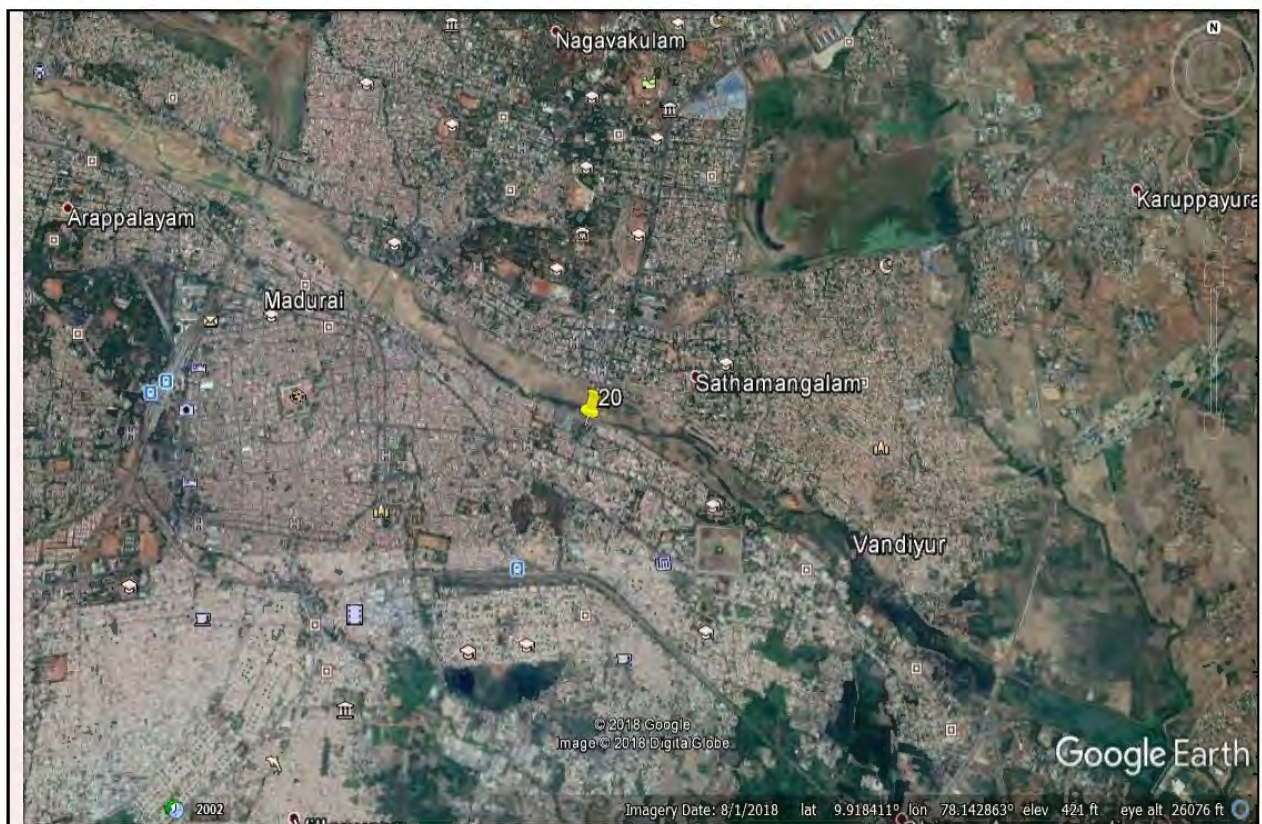
Location No: 31 (13) - Kochadai back side of Pumping station



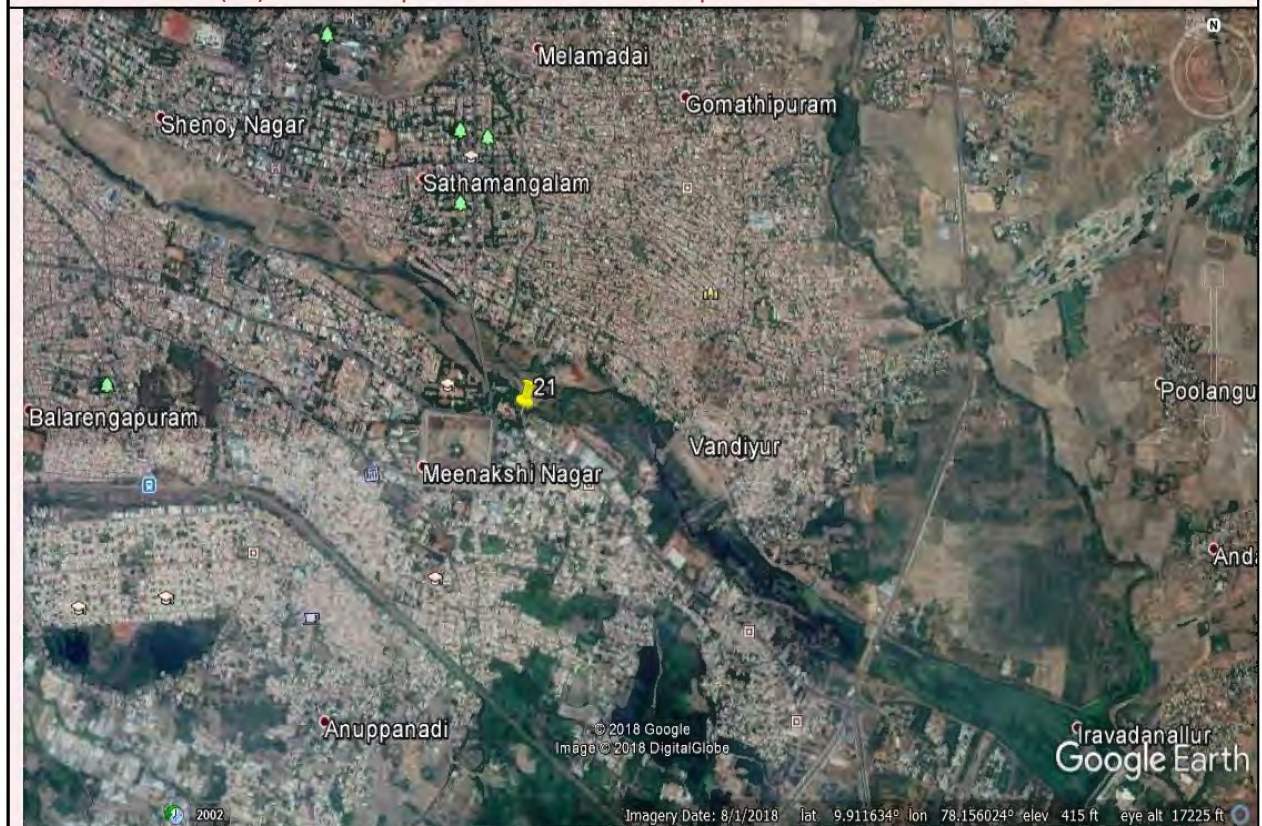
Location No: 32 (12) - VKP Nagar Madakulam



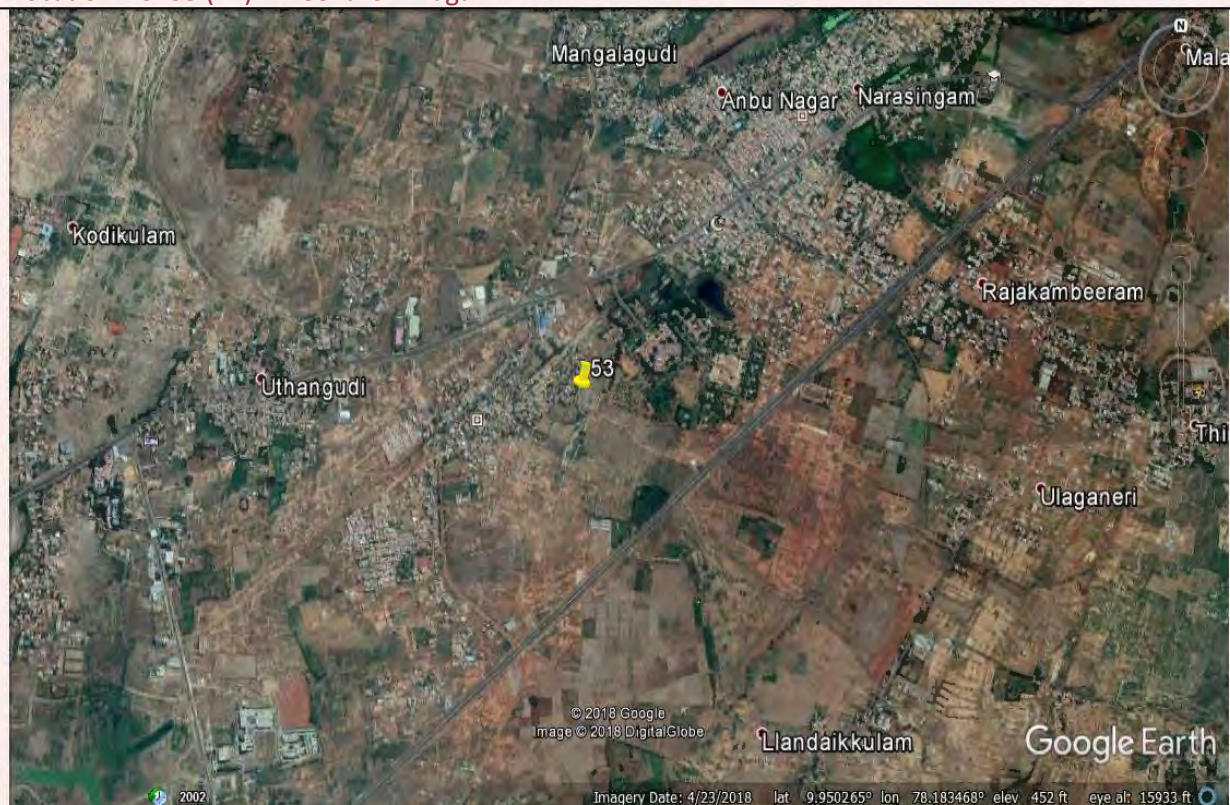
Location No: 33 (10) - Arappalayam Two wheeler Parking



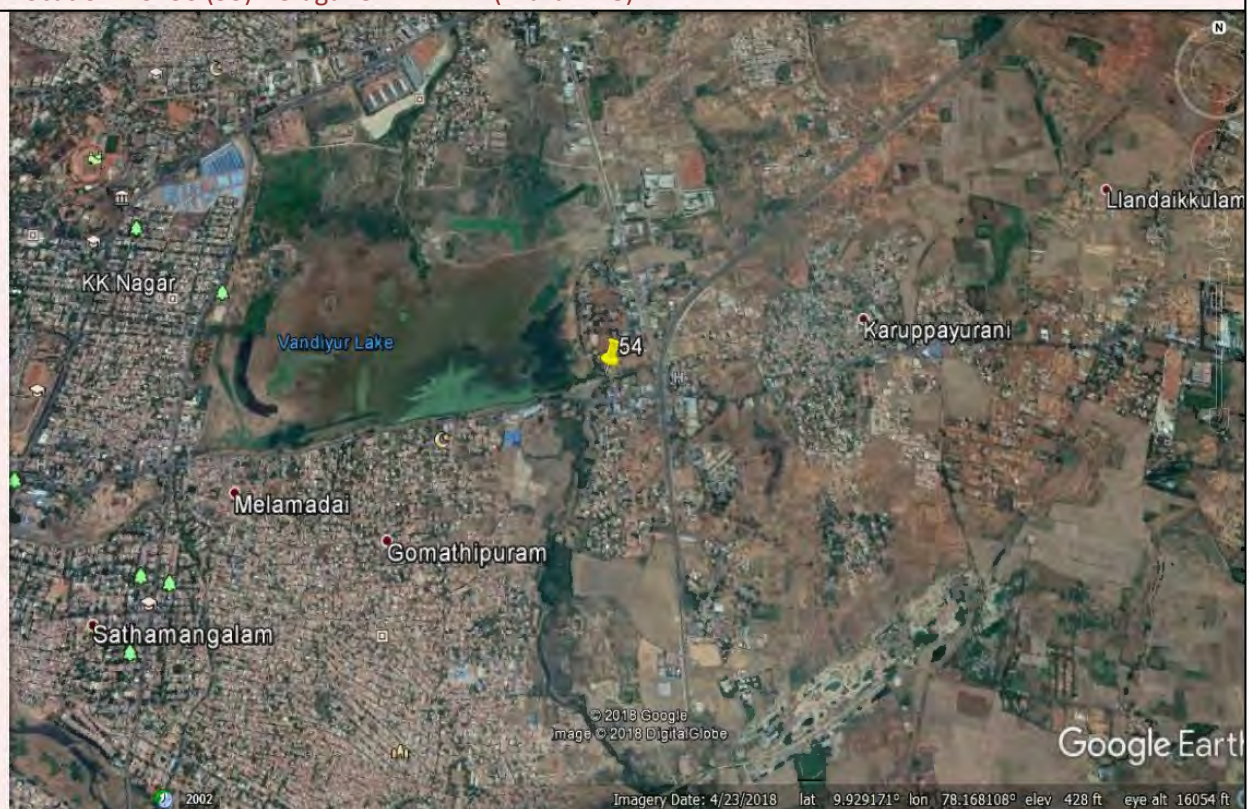
Location No: 34 (20) - AVSS Hospital behind MPS Santhaipettai



Location No: 35 (21) - Meenakshi Nagar



Location No: 36 (53) - Ulaganeri – NZ 11 (Ward – 28)



Location No: 37 (54) - Pandian Nagar – NZ12

Appendix- 16

Public Consultation at Overhead tank locations

A total of 38 locations (37 OHTs, 1 Sump) has been identified for water distribution under the **“Dedicated Water Supply Scheme for Madurai Municipal Corporation from Mullai Periyar at Lower camp”**. Out of 38 OHT's, 14 OHT locations were identified for conducting public consultations, which was held from 13 February 2019 to 15 February 2019. The locations were selected based on the presence of socially important locations including the public park area, playground, burial ground, temples etc., consultation was held. The outcome of the consultation has been discussed in the following tables.

I. Minutes of meeting for “Dedicated Water Supply Scheme for Madurai Municipal Corporation from Mullai Periyar at Lower Camp” held on 13 February 2019

S. No	Participants	Outcome of the Consultation and Discussions	Response from PIU	Photographs
1.	OHT # 14 and 15 OHT Location: TVS Park Sathyasai Nagar M. Maheshwari, K. Marreswari, M. Barani, M. Sweetey, R. Ghanam, K. Selvarajan, L. Alagiriraj, P.S. Pandiyan, S.G. Palsamy, K. Alagarsamy, T.K.S. Omana, N.V. Dhayarathi – HM (TVS - HSS) P. Jeyanthi – HM (TVS - HSS)	In general, the proposal has been welcomed by the local people and they have shown unanimous support for the project - There is no objection for the construction of the OHT in the Public Park. - What would be the project completion date?	It was informed by the PIU, that the work shall be completed within two years from the date of contract	
2.	OHT# 47 OHT location: Vellakal Rasuthevar, Pandi, M. Karunanithi, P. Arunachalam, T. Rajkumar, P. Prabhu, S. Renugadevi, J. Pasumpon, R. Jeyanthimala, P.	It was informed that the area is facing irregular water supply and hence it was requested to construct the OHT at the earliest so as to get regular	It was assured by the PIU to construct the OHT at the earliest so as to facilitate the regular water	

S. No	Participants	Outcome of the Consultation and Discussions	Response from PIU	Photographs
	Malliga, A. Meenakshi, Murugayi, Chitra Ponnammal, Petchi, Pandiyammal, Reyalakshmi, T. Shanmuganathan	water	supply	
3.	OHT# 48 OHT location: Harvipatti S. Mariappan, G. Elankumaran, M. Ayyavu, K. Murugan, S. Ramalakshmi, K. Muthu, K. Amutha, P. Ellappan, Ganguly, G. Ramakrishnan, A. Jayabalan, M. Sangaiya, L. Raja murugan, P. Saravanapandi, S. Babu, G. Magudapathy, S. Pathiban (VIII A), G. Chandran (VIII A), R. Ramar, R. Lakshmanan, S. Dharun, K. Manikandan, S. Sanjay Kumar, S. Vetrivelan, R. Harivignesh, R.P. JeevaRagavan	It was informed that due to irregular water supply, there is scarcity of water for drinking purposes and hence the local people have requested to construct the OHT at the earliest. However, it was suggested to shift the location of the OHT near to an existing one, so as to provide space for the children to use it as a play area. It was requested to have proper barricading/ fence to prevent children's entering the OHT area. It was suggested not to have borewell in the area	The PIU have accepted the request and assured that the new OHT shall be constructed near the existing one and it will be properly barricaded to prevent the children's entering the site. It was assured that there borewell will not be constructed in the OHT area	

S. N o	Participants	Outcome of the Consultation and Discussions	Response from PIU	Photographs
4.	OHT# 50 OHT location: Balaji Nagar A.A. Sethuram, P. Rajendran, V. Ganapathi, N. Shanmugasundaram, K. Palanikumar, A.J. Abdul nazeer, R. Amuthavalli, T. Murugan, V. Ganesh, G. Janarthanan	Similar to Harvipatti area, due to irregular water supply, the regular activities are getting hampered. Hence, the local people have requested the PIU to construct the OHT at the earliest to solve the water related issues and also they have requested the PIU to shift the OHT inside the public park	The PIU have accepted the request and assured that the OHT shall be located inside the Public park	

Minutes of meeting for “Dedicated Water Supply Scheme for Madurai Municipal Corporation from Mullai Periyar at Lower Camp” held on 14 February 2019

S. No	Participants	Outcome of the Consultation and Discussions	Response from PIU	Photographs
1.	OHT# 19 OHT location: Vaalaithoppu Corporation Ground G. Kumar, Arumugam, V. Jayarani, Pandiamma, A. Kalieeswari, M. Pooppandi, R. Senthorpand, A. Kalleswari, N. Katturaja and S. Murugan	The local community have requested the PIU to construct the OHT and to provide water supply at the earliest		
2.	OHT# 21 OHT location: Meenakshi Nagar L. Nagaraj, S. Vargeeshbabu, S. Kannan, Shanthi, Mary, M. Savarimuthu, Lakshmanan, A. Andisami and Kuruvammal	The local community have requested the PIU to construct the OHT and to provide water supply at the earliest For the PAP's who are losing the structures are requested for appropriate compensation .	It was assured by the PIU that the PAP's shall be compensated as per the provisions given in the RP	
3.	OHT# 36 OHT location: Island Nagar M. Ramu, C. Vetrivel, M. Navaneeth, M. Ramkumar, PA. Muthupandi, S. Balamurugan, C. Dhayalammal, D. Jenifer, R.	One of the PAP have stated that he is a fisherman depending on the nearby water tank for fishing. However, due to the scarcity of water in the tank, he	PIU have informed that notice has been issued well in advance for vacating the place. For the loss of the structure, compensation has been	

S. No	Participants	Outcome of the Consultation and Discussions	Response from PIU	Photographs
	Rathinam, S. Yasmai and Thangam	opted for an alternate job (mosaic polishing). Due to his absence, his house was demolished without prior information/ notice. Few of the local community members have requested for an approach leading to the nearby temple.	provisioned in the RP, PIU have assured that the PAP shall be compensated as per the provisions of RP. The PIU have assured that 2 m width approach road to the temple shall be provided for the benefit of the local people.	
4.	OHT# 38 OHT location: Shenpagathottam A. Varusaimohamed, A. Ninalan and J. Gabriel	The local community have requested the PIU to construct the OHT and to provide water supply at the earliest		
5.	OHT# 41 OHT location: ChinnaAnuppanadi K. Abimani, Vinoth, S. Santhosh, Kumar, N. Amir abbas, R. Vasanth, K. Kandavelpandian, D. Karthikeyan, M. Sathasivam, K. J. Lalitha, T.G. Savithiri, Chellammal, N.	The local community have no objection to the construction of the OHT. However, they have requested the PIU to construct the OHT in such a way not to disturb the children's	It was assured by the PIU that the playing activity will not be disturbed during the construction activities. The PIU have clarified that there is no provision to have	

S. N o	Participants	Outcome of the Consultation and Discussions	Response from PIU	Photographs
	Uma, E. Kiruba, M. Kannan, S. SanthanaShanthi, I. Sahayarajan and S. Suresh	playing activity. Few have also commented that they don't want to have borewell in that location	borewell in this project. The water shall be sourced from Mullaiperiyar dam (not groundwater)	

Minutes of meeting for “Dedicated Water Supply Scheme for Madurai Municipal Corporation from MullaiPeriyar at Lower Camp” held on 15 February 2019

S. N o	Participants	Outcome of the Consultation and Discussions	Response from PIU	Photographs
1.	OHT# 42 OHT location: GurunatharKoil – Chinnthamani S. Chitra, S. Vijayalakshmi, T. Mary Joyce, P. PandiSelvan, V. SethuMadhavan, S. Ganeshan and V. Ram	The local community have supported the project and requested the PIU to provide compound wall surrounding the OHT for safety purposes.	PIU have assured to provide compound wall / fencing surrounding the OHT.	
2.	OHT# 25 OHT location: Semparuthi Nagar C. Prasath, P. Jeyaprakash, P. Nagajothi, S. Shanthi, K. Maheswari, A. Gayathiri, A. Sakthivel, Sam George, M. Hemalatha, R. priya, R. Raja, M. Gurusamy, R. Lakshmi and K. Muthulakshmi	The local community have supported the project and requested the PIU to construct the OHT.		
3.	OHT# 32 OHT location: EB Colony P. Manimaran, P. Premalatha, S. Steeladevi, Arammal, S. Senthilvel, V. Pirathaban, Karuppasamy, P. Arivalagan, T. Jeyakodi and P. Madasamy	The local community have supported the project and have requested the PIU to construct the OHT.		

S. No	Participants	Outcome of the Consultation and Discussions	Response from PIU	Photographs
4.	OHT# 34 OHT location: Bharath Nagar P. Muthuraman, P. Moorthy, J. Hasanbanu and A. Chellama	The local community have supported the project and requested the PIU to construct the OHT.		
5.	OHT# 37 OHT location: Poriyal Nagar Naveen Yasav, M.A. Shanmugam, K. Karuppiyah and S. Vijayalakshmi	The local community have supported the project and requested the PIU to construct the OHT. However, the community have raised a query regarding the technology used in the construction of the OHT. Because, due to the soil instability one of the house got damaged and it was recently renovated.	PIU have clarified the query with respect to the technology used for the construction of the OHT.	

APPENDIX 17

Public Works department No Objection Certificate for Headworks

Appendix-18

Monthly Average Storage in Mullai Periyar Dam (january 2012 to Dec 2023)

Year
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023

நீர்வளத்துறை

அனுப்புநர்
திரு. ஜே. சாம் இர்வின், பி.இ.,
கண்காணிப்புப்பொறியாளர், நீவது,
செயற்பொறியாளர், நீவது, (சு.பொ)
பெரியாறு அணை சிறப்புக் கோட்டம்,
கம்பம் - 625 516.

பெறுநர்
தலைமை பொறியாளர்,
மதுரை மாநகராட்சி,
அறிஞர் அண்ணா மாளிகை,
தல்லாகுளம்,
மதுரை-625002

க. எண்: 85எம்/வப/உ.பொ.1/கோ.43/2024/நாள்: 13.08.2024

ஐயா,

பொருள்: முல்லை பெரியாறு அணையின் நீர்த்தேக்க விவரங்கள் வழங்குதல் -
தொடர்பாக.

பார்வை: தலைமை பொறியாளர், மதுரை மாநகராட்சி, அறிஞர் அண்ணா
மாளிகை, தல்லாகுளம், மதுரை அவர்களின் க.எண். ஜே.2/அம்ருத்
(குடிநீர் திட்டம்)/2018/நாள்: 12.08.2024

*** **

. WRD,
il Division,

Dec
0 1,683.00
0 2,119.00
0 5,488.00
0 7,330.00
0 1,177.00
0 3,914.00
0 4,099.00
3 4,105.97
23 3,305.84
30 7,622.58
53 7,403.68
53 6,504.39

மதுரை மாநகராட்சிக்கு முல்லை பெரியாற்றின் லோயர் கேம்பு பகுதியிலிருந்து
நீர் எடுத்து சுத்திகரிப்பு செய்து குடிநீர் வழங்கும் திட்டம் சம்பந்தமாக, ஆகிய வளர்ச்சி
வங்கிக்கு சமர்ப்பிக்கும் பொருட்டு முல்லை பெரியாறு அணையின் கொள்ளளவு
தொடர்பான தரவுகளை வழங்குமாறு பார்வையில் காணும் கடிதத்தில்
கேட்டுக்கொள்ளப்பட்டது.

எனவே, 1. 2019ம் வருடம் முதல் 2023 வரை மாதம் வாரியாக முல்லை பெரியாறு
அணையின் நீர்த்தேக்க சராசரி கொள்ளளவு

2. 2019 முதல் மாதாந்திர கொள்ளளவின் சுருக்கம் ஆகிய தரவுகளை இத்துடன்
இணைத்து அனுப்பப்படுகிறது.

இணைப்பு: படிவம்-2 எண்ணம்

செயல்பொறியாளர், நீவது, கம்பம்
பெரியாறு அணை சிறப்புக் கோட்டம், கம்பம்

நீர்வளத்துறை

அனுப்புநர்
திரு. ஜே. சாம் இர்வின், பி.இ.,
கண்காணிப்புப்பொறியாளர், நீவது,
செயற்பொறியாளர், நீவது, (சு.பொ)
பெரியாறு அணை சிறப்புக் கோட்டம்,
கம்பம் - 625 516.

பெறுநர்
தலைமை பொறியாளர்,
மதுரை மாநகராட்சி,
அறிஞர் அண்ணா மாளிகை,
தல்லாகுளம்,
மதுரை-625002

க. எண்: 85எம்/வப/உபொ.1/கோ.43/2024/நாள்: 13.08.2024

ஐயா,

பொருள்: முல்லை பெரியாறு அணையின் நீர்த்தேக்க விவரங்கள் வழங்குதல் -
தொடர்பாக.

பார்வை: தலைமை பொறியாளர், மதுரை மாநகராட்சி, அறிஞர் அண்ணா
மாளிகை, தல்லாகுளம், மதுரை அவர்களின் க.எண். ஜே.2/அம்ருத்
(குடிநீர் திட்டம்)/2018/நாள்: 12.08.2024

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மதுரை மாநகராட்சிக்கு முல்லை பெரியாற்றின் லோயர் கேம்ப் பகுதியிலிருந்து
நீர் எடுத்து சுத்திகரிப்பு செய்து குடிநீர் வழங்கும் திட்டம் சம்பந்தமாக, ஆசிய வளர்ச்சி
வங்கிக்கு சமர்ப்பிக்கும் பொருட்டு முல்லை பெரியாறு அணையின் கொள்ளளவு
தொடர்பான தரவுகளை வழங்குமாறு பார்வையில் காணும் கடிதத்தில்
கேட்டுக்கொள்ளப்பட்டது.

எனவே, 1. 2019ம் வருடம் முதல் 2023 வரை மாதம் வாரியாக முல்லை பெரியாறு
அணையின் நீர்த்தேக்க சராசரி கொள்ளளவு

2. 2019 முதல் மாதாந்திர கொள்ளளவின் சுருக்கம் ஆகிய தரவுகளை இத்துடன்
இணைத்து அனுப்பப்படுகிறது.

இணைப்பு: படிவம்-2 எண்ணம்

P. R. 13/8/24
செயற்பொறியாளர், நீவது, *செயல்பு*
பெரியாறு அணை சிறப்புக் கோட்டம், கம்பம்

**Table 6: Abstract of Monthly Storage of Mullaperiyar Dam
January 2012 to December 2023**

No.	Year	Monthly Storage (MCFT)		
		Minimum	Maximum	Average
1	2012	810.48	2,366.58	1,489.35
2	2013	1025.94	5,134.61	2,272.91
3	2014	1,021.42	6,788.13	2,889.58
4	2015	1168.13	7,330.16	2,796.46
5	2016	911.9	5,043.77	1,746.77
6	2017	755.23	3,914.23	1,743.98
7	2018	1,316.20	6,477.84	3,384.59
8	2019	1,251.80	4,303.50	2,603.62
9	2020	1,304.17	4,870.71	2,710.33
10	2021	3,980.10	7,622.58	5,165.47
11	2022	3,847.13	7,403.68	5,467.23
12	2023	1,947.50	6,504.39	3,336.51
Total		19,340.00	67,760.17	35,606.80
Monthly Average in Mcft		1,611.67	5,646.68	2,967.23

for *P.A. 13/8/24*
Executive Engineer, WRD,
Periyar Dam Special Division,
Cumbum - 625516

Appendix-19

Pre-construction Environmental test report (Package 1, 2, 3 & 4)

Package-1 Baseline Environmental test Report (Water, Air and Noise test Report)

TEST REPORT		 Enviro Care[®] INDIA PRIVATE LIMITED ISO 9001 : 2008 Certified Company ISO 14001 : 2004 Certified Company OHSAS : 18001 : 2007 Certified Company Accredited by NABL (Environmental & Biological) Reference No. 117/2015, as per IS : 14543 : 2014
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WATER ANALYSIS

Report No :	ECI-NN-VVA-128/07/2020	Report Date :	27.07.2020	
Customer Name & Address	M/s. The Indian Hume Pipe co Ltd, Second Floor B-2, Le Royal Ritter Roof, New No.361Lyods Road, Gopalapuram Chennai-600 08.			
Customer Reference :	IWO DL :21/07/2020	Sample Reference No :	ECI-NN-WA-128/07/2020	
Sample Drawn By :	Customer	Sample Received On :	22.07.2020	
Sample Collected Date :	21.07.2020	Test Commenced On :	22.07.2020	
Qty. of Sample Received :	2 L.	Test Completed On :	27.07.2020	
Sample Description :	Water	Sampling Method :		
Sample Mark:	Ground VWater- Near Vaigaidam Gate			
S.No	PARAMETERS	UNITS	RESULTS	TEST METHOD
1.	Bio chemical oxygen demand (BOD @ 27°C for 3 days)	mg/L	<1.0	IS 3025:Part 44
2.	Calcium(as Ca)	mg/L	113	IS 3025:Part 40
3.	Chemical Oxygen Demand (COD)	mg/L	<1.0	IS 3025:Part 58
4.	Chlorides(as Cl)	mg/L	467	IS 3025:Part 32
5.	Color	Hazen	<1.0	IS 3025:Part 04
6.	Cyanide(as CN)	mg/L	< 0.02	IS 3025:Part 27
7.	Dissolved Oxygen	mg/L	5.5	IS 3025:Part 38
8.	Fluoride (as F)	mg/L	< 0.1	IS 3025:Part 60
9.	Iron (as Fe)	mg/L	BDL (DL = 0.01)	IS 3025:Part 53
10.	Magnesium (as Mg)	mg/L	36	IS 3025:Part 46
11.	Nitrite Nitrogen(as NO ₂)	mg/L	BDL (DL = 0.01)	IS 3025:Part 34
12.	pH Value @ 25°C	—	7.0	IS 3025:Part 11
13.	Phenolic Compound (as C ₆ H ₅ OH)	mg/L	< 0.001	IS 3025:Part 43
14.	Potassium (as K)	mg/L	34	IS 3025:Part 45
15.	Reactive Silica (as SiO ₂)	mg/L	18.3	IS 3025:Part 35
16.	Residual Free Chlorine (RFC)	mg/L	< 0.1	IS 3025:Part 26
17.	Sodium(as Na)	mg/L	81	IS 3025:Part 45
18.	Sulphates (as SO ₄ ²⁻)	mg/L	159	IS 3025:Part 24
19.	Total Alkalinity (as CaCO ₃)	mg/L	302	IS 3025:Part 23
20.	Total Chromium (as Cr)	mg/L	BDL (DL = 0.01)	IS 3025:Part 52

NOTE :

- The results relate only to the items listed.
- Any material found above the stated limit must be reported.
- None of the test results are to be used for any purpose without the permission of the laboratory.
- Unless otherwise specified, the test results are valid for 15 days from date of report.
- Total quantity of our laboratory is limited to the capacity of the equipment.
- Any dispute arising out of this report is referred to the nearest court of law.

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TEST REPORT		 Enviro Care[®] INDIA PRIVATE LIMITED ISO 9001 : 2008 Certified Company ISO 14001 : 2004 Certified Company OHSAS : 18001 : 2007 Certified Company Accredited by NABL (Environmental & Biological) Reference No. 117/2015, as per IS : 14543 : 2014
--------------------	--	--

Report No : ECI-NN-WA-128/07/2020				
21.	Total Dissolved Solids(TDS) @ 105°C	mg/L	1165	IS 3025:Part 16
22.	Total Hardness(as CaCO ₃)	mg/L	361	IS 3025:Part 21
23.	Total Suspended Solids(TSS) @105°C	mg/L	<1.0	IS 3025:Part 17
24.	Turbidity	NTU	< 0.5	IS 3025:Part 10
<--- End of Report --->				
Verified By : <u>S. K. Srinivas Reddy</u> Remarks : BDL - Below Detectable Limit, DL - Detectable Limit For ENVIRO CARE INDIA PRIVATE LIMITED (Laboratory Division) Authorized Signatory				



WATER ANALYSIS

Report No :	ECI-NN-WA-313/06/2020	Report Date :	29.06.2020	
Customer Name & Address	M/s. The Indian Hume Pipe co Ltd, Second Floor B-2, Le Royal Ritter Roof, New No.361Llyods Road Gopalapuram Chennai-600 08			
Customer Reference :	IWO Dt:25/06/2020	Sample Reference No :	ECI-NN-WA-313/06/2020	
Sample Drawn By :	Customer	Sample Received On :	26.06.2020	
Sample Collected Date :	25.06.2020	Test Commenced On :	26.06.2020	
Qty of Sample Received :	2 L	Test Completed On :	29.06.2020	
Sample Description :	Water	Sampling Method :	--	
Sample Mark:	Ground Water- K.Vilakku (NE)			
S.No	PARAMETERS	UNITS	RESULTS	TEST METHOD
1	Bio chemical oxygen demand (BOD @ 27°C for 3 days)	mg/L	<1.0	IS 3025 Part 44
2	Calcium(as Ca)	mg/L	139	IS 3025 Part 40
3	Chemical Oxygen Demand (COD)	mg/L	<1.0	IS 3025 Part 58
4	Chlorides(as Cl)	mg/L	526	IS 3025 Part 32
5	Color	Hazen	<1.0	IS 3025 Part 04
6	Cyanide(as CN)	mg/L	< 0.02	IS 3025 Part 27
7	Dissolved Oxygen	mg/L	6.1	IS 3025 Part 38
8	Fluoride (as F)	mg/L	< 0.1	IS 3025 Part 60
9	Iron (as Fe)	mg/L	BDL (DL= 0.01)	IS 3025 Part 53
10	Magnesium (as Mg)	mg/L	47	IS 3025 Part 46
11	Nitrite Nitrogen(as NO ₂ -)	mg/L	BDL (DL = 0.01)	IS 3025 Part 34
12	pH Value @ 25°C	--	7.2	IS 3025 Part 11
13	Phenolic Compound (as C ₆ H ₅ OH)	mg/L	< 0.001	IS 3025 Part 43
14	Potassium (as K)	mg/L	45	IS 3025 Part 45
15	Reactive Silica (as SiO ₂ -)	mg/L	25.6	IS 3025 Part 35
16	Residual Free Chlorine (RFC)	mg/l	< 0.1	IS 3025 Part 26

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- Note :**
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 2. Any correction not attested shall invalidate this report.
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 6. Any dispute arising out of this report is subjected to Madurai Jurisdiction Only.





Accredited by **NABL** (Chemical & Biological)
Recognized by BIS as per IS : 14543 : 2016

Report No : ECI-NN-WA-313/06/2020				
17	Sodium(as Na)	mg/L	102	IS 3025 :Part 45
18	Sulphates (as SO_4^{2-})	mg/L	242	IS 3025 Part 24
19	Total Alkalinity (as CaCO_3)	mg/L	343	IS 3025 Part 23
20	Total Chromium (as Cr)	mg/L	BDL (DL = 0.01)	IS 3025 Part 52
21	Total Dissolved Solids(TDS) @ 105°C	mg/L	1352	IS 3025:Part 16
22	Total Hardness(as CaCO_3)	mg/L	463	IS 3025 Part 21
23	Total Suspended Solids(TSS) @ 105°C	mg/L	<1.0	IS 3025 Part 17
24	Turbidity	NTU	< 0.5	IS 3025:Part 10

<--- End of Report --->

Verified By :

Remarks : BDL –Below Detectable Limit, DL – Detectable Limit

For ENVIRO CARE INDIA PRIVATE LIMITED
(Laboratory Division)

Authorized Signatory



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TEST REPORT
 Accredited by **NABL** (Chemical & Biological)
 Recognized by **BIS** as per IS : 14543 : 2016
AMBIENT AIR QUALITY SURVEY

Report No :	ECI-NN-AAQ-128/07/2020	Report Date :	27.07.2020
Customer Name & Address	M/s. The Indian Hume Pipe co Ltd, Second Floor B-2, Le Royal Ritter Roof, New No:361Llyods Road, Gopalapuram Chennai-600 086		
Customer Reference :	IWO Dt: 21/07/2020	Sample Reference No :	ECI-NN-AAQ-128/07/2020
Sample Drawn By :	ECI	Sample Received On :	22.07.2020
Sample Collected Date :	21.07.2020	Test Commenced On :	22.07.2020
Qty of Sample Received :	Filter Paper & 25ml Solution	Test Completed On :	27.07.2020
Sample Description :	Ambient Air	Sampling Method :	IS 5182:P14
Sample Mark:	Near Vaigaidam Gate		

S.No	PARAMETERS	UNITS	RESULTS	TEST METHOD	Permissible limits of NAAQs (Industrial, Residential)
1.	Carbon Monoxide (as CO)	mg/m ³	< 1.2	ECI-SOP-SAM-08	2.0
2.	Lead (as Pb)	µg/m ³	< 0.1	IS 5182:Part 22	1.0
3.	Nitrogen dioxide (as NO ₂)	µg/m ³	16.8	IS 5182:Part 06	80
4.	Respirable Particulate Matter (PM _{2.5})	µg/m ³	41.2	IS 5182:Part 23	60
5.	Sulphur Dioxide (as SO ₂)	µg/m ³	9.1	IS 5182:Part 02	80
6.	Suspended Particulate Matter (PM ₁₀)	µg/m ³	94.3	IS:5182 Part 4	100

<--- End of Report --->

Verified By : *[Signature]*

Remarks : --


For ENVIRO CARE INDIA PRIVATE LIMITED
 (Laboratory Division)
[Signature]
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AMBIENT AIR QUALITY SURVEY

Report No :	ECI-NN-AAQ-295/06/2020	Report Date :	29.06.2020
Customer Name & Address	M/s. The Indian Hume Pipe co Ltd, Second Floor B-2, Le Royal Ritter Roof, New No.361Llyods Road, Gopalapuram Chennai-600 086		
Customer Reference :	IWO Dt:25/06/2020	Sample Reference No :	ECI-NN-AAQ-295/06/2020
Sample Drawn By :	ECI	Sample Received On :	26.06.2020
Sample Collected Date :	25.06.2020	Test Commenced On :	26.06.2020
Qty of Sample Received :	Filter Paper & 25ml Solution	Test Completed On :	29.06.2020
Sample Description :	Ambient Air	Sampling Method :	IS 5182-P 14
Sample Mark:	K Vilaku (NE)		

S.No	PARAMETERS	UNITS	RESULTS	TEST METHOD	Permissible limits of NAAQs (Industrial, Residential)
1.	Carbon Monoxide (as CO)	mg/m ³	< 1.2	ECI-SOP-SAM-08	2.0
2.	Lead (as Pb)	µg/m ³	< 0.1	IS 5182 Part 22	1.0
3.	Nitrogen dioxide (as NO ₂)	µg/m ³	10.8	IS 5182 Part 06	80
4.	Respirable Particulate Matter (PM _{2.5})	µg/m ³	29.5	IS 5182 Part 23	60
5.	Sulphur Dioxide (as SO ₂)	µg/m ³	6.2	IS 5182 Part 02	80
6.	Suspended Particulate Matter (PM ₁₀)	µg/m ³	75.2	IS 5182 Part 4	100

< --- End of Report --- >

Verified By :	
Remarks :	--

For ENVIRO CARE INDIA PRIVATE LIMITED
(Laboratory Division)

Authorized Signatory

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AMBIENT AIR QUALITY SURVEY

Report No :	ECI-NN-AAQ-296/06/2020	Report Date :	29.06.2020
Customer Name & Address	M/s. The Indian Hume Pipe co Ltd, Second Floor B-2, Le Royal Ritter Roof, New No.361Llyods Road, Gopalapuram Chennai-600 086		
Customer Reference :	IWO Dt:25/06/2020	Sample Reference No :	ECI-NN-AAQ-296/06/2020
Sample Drawn By :	ECI	Sample Received On :	26.06.2020
Sample Collected Date :	25.06.2020	Test Commenced On :	26.06.2020
Qty of Sample Received :	Filter Paper & 25ml Solution	Test Completed On :	29.06.2020
Sample Description :	Ambient Air	Sampling Method :	IS 5182-P14
Sample Mark:	Veerapandi (SE)		

S.No	PARAMETERS	UNITS	RESULTS	TEST METHOD	Permissible limits of NAAQs (Industrial, Residential)
1	Carbon Monoxide (as CO)	mg/m ³	< 1.2	ECI-SOP-SAM-08	2.0
2	Lead (as Pb)	µg/m ³	< 0.1	IS 5182 Part 22	1.0
3	Nitrogen dioxide (as NO ₂)	µg/m ³	10.5	IS 5182 Part 06	80
4	Respirable Particulate Matter (PM _{2.5})	µg/m ³	25.4	IS 5182 Part 23	60
5	Sulphur Dioxide (as SO ₂)	µg/m ³	5.9	IS 5182 Part 02	80
6	Suspended Particulate Matter (PM ₁₀)	µg/m ³	65.3	IS 5182 Part 4	100

<--- End of Report --->

Verified By :	For ENVIRO CARE INDIA PRIVATE LIMITED (Laboratory Division) Authorized Signatory
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Note: 1. The results relate only to this item tested.

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Recognized by BIS as per IS : 14543 : 2016

AMBIENT AIR QUALITY SURVEY

Report No :	ECI-NN-AAQ-298/06/2020	Report Date :	29.06.2020
Customer Name & Address	M/s. The Indian Hume Pipe co Ltd, Second Floor B-2, Le Royal Ritter Roof, New No.361Llyods Road, Gopalapuram Chennai-600 086		
Customer Reference :	IWO Dt 25/06/2020	Sample Reference No :	ECI-NN-AAQ-298/06/2020
Sample Drawn By :	ECI	Sample Received On :	26.06.2020
Sample Collected Date :	25.06.2020	Test Commenced On :	26.06.2020
Qty of Sample Received :	Filter Paper & 25ml Solution	Test Completed On :	29.06.2020
Report No :	ECI-NN-AAQ-296/06/2020	Report Date :	29.06.2020
Sample Mark:	Lower Camp (NW)		

S.No	PARAMETERS	UNITS	RESULTS	TEST METHOD	Permissible limits of NAAQs (Industrial, Residential)
1.	Carbon Monoxide (as CO)	mg/m ³	< 1.2	ECI-SOP-SAM-08	2.0
2.	Lead (as Pb)	µg/m ³	< 0.1	IS 5182 Part 22	1.0
3.	Nitrogen dioxide (as NO ₂)	µg/m ³	9.6	IS 5182 Part 06	80
4.	Respirable Particulate Matter (PM _{2.5})	µg/m ³	28.8	IS 5182 Part 23	60
5.	Sulphur Dioxide (as SO ₂)	µg/m ³	4.9	IS 5182 Part 02	80
6.	Suspended Particulate Matter (PM ₁₀)	µg/m ³	72.6	IS 5182 Part 4	100

<--- End of Report --->

Verified By :	For ENVIRO CARE INDIA PRIVATE LIMITED (Laboratory Division) Authorized Signatory
Remarks : --	



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Note : The results relate only to this item tested.

- Any correction not attested shall invalidate this report.
- Report shall not be reproduced anywhere except in full and in the same format without the permission of the laboratory.
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Accredited by **NABL** (Chemical & Biological)
Recognized by BIS as per IS : 14543 : 2016

NOISE MONITORING

Report No :	ECI-NN-NM-258/06/2020	Report Date :	29.06.2020
Customer Name & Address	M/s. The Indian Hume Pipe co Ltd, Second Floor B-2, Le Royal Ritter Roof, New No.3611Lyoris Road, Gopalapuram Chennai-600 086		
Customer Reference :	IWO Dt.25/06/2020	Sample Reference No :	ECI-NN-NM-258/06/2020
Sample Description :	Noise	Monitoring Date :	25.06.2020
Monitored By :	ECI	Data Received On :	26.06.2020

S.No	Locations	Day Time	
		Minimum	Maximum
1	K Villaku (NE)	65.3	67.9
2	Veerapandi (SE)	61.2	62.3
3	Uttamapalayam (SW)	63.8	64.7
4	Lower Camp (NW)	59.3	61.5
Unit		dB (A)	
TNPCB Max. Permissible Limit for Day Time (Industrial Area)		75	
Reference Method		IS 9989	

<--- End of Report --->

Verified By :	For ENVIRO CARE INDIA PRIVATE LIMITED (Laboratory Division) Authorized Signatory
----------------------	--

Remarks: In the above mentioned locations meets the requirements of TNPCB standards



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Note : 1. The results relate only to this item tested.

2. Any correction not attested shall invalidate this report.

3. Report shall not be reproduced anywhere except in full and in the same format without the permission of the laboratory.

4. Unless informed by customer, the test items will not be retained for more than 15 days from date of issue of test report.

5. Total liability of our laboratory is limited to the invoice amount.

6. Any dispute arising out of this report is subjected to Madurai Jurisdiction Only.



TEST REPORT
 Accredited by **NABL** (Chemical & Biological)
 Recognized by BIS as per IS : 14543 : 2015
NOISE MONITORING

Report No :	ECI-NN-NM-94/07/2020	Report Date :	27.07.2020
Customer Name & Address	M/s. The Indian Hume Pipe co Ltd, Second Floor B-2, Le Royal Ritter Roof, New No:361Llyods Road, Gopalapuram Chennai-600 086		
Customer Reference :	IWO Dt: 21/07/2020	Sample Reference No :	ECI-NN-NM-94/07/2020
Sample Description :	Noise	Monitoring Date :	21.07.2020
Monitored By :	ECI	Data Received On :	22.07.2020

S.No	Locations	Day Time	
		Minimum	Maximum
1.	Near Vaigaidam Gate	68.6	70.1
Unit		dB (A)	
TNPSB Max. Permissible Limit for Day Time (Industrial Area)		75	
Reference Method		IS 9989	

<--- End of Report --->

Verified By : *S. Mahalingam*

Remarks: In the above mentioned locations meets the requirements of TNPSB standards.

For ENVIRO CARE INDIA PRIVATE LIMITED
(Laboratory Division)
[Signature]
Authorized Signatory

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Package-2 Baseline Environmental test Report (Air, Noise and Water test Report)

**TEST REPORT**

Accredited by **NABL** (Chemical & Biological)
Recognized by BIS as per IS : 14543 : 2016

AMBIENT AIR QUALITY SURVEY

Report No :	ECI-NN-AAQ-174/10/2020	Report Date :	19.10.2020
Customer Name & Address	M/s. P&C Projects (P) Ltd., WTP Project Site Pannaipatti, Viralipatti, Nilakottai Taluk, Battalagundu, Dindigul Dist.		
Customer Reference :	IWO Dt: 14/10/2020	Sample Reference No :	ECI-NN-AAQ-174/10/2020
Sample Drawn By :	ECI	Sample Received On :	15.10.2020
Sample Collected Date :	14.10.2020	Test Commenced On :	15.10.2020
Qty of Sample Received :	Filter Paper & 25ml Solution	Test Completed On :	19.10.2020
Sample Description :	Ambient Air	Sampling Method :	IS 5182: P14
Latitude :	10.1135325 N	Longitude :	77.7353723 E
Sample Mark:	Adjacent Area Within 600 m Eastern Side Of WTP-(Mr. Subbiah Garden)		

S.No	PARAMETERS	UNITS	RESULTS	TEST METHOD	Permissible limits of NAAQs (Industrial, Residential)
1.	Carbon Monoxide (as CO)	mg/m ³	< 1.2	ECI-SOP-SAM-08	2.0
2.	Lead (as Pb)	µg/m ³	< 0.1	IS 5182: Part 22	1.0
3.	Nitrogen dioxide (as NO ₂)	µg/m ³	7.9	IS 5182: Part 06	80
4.	Respirable Particulate Matter (PM _{2.5})	µg/m ³	15.0	IS 5182: Part 23	60
5.	Sulphur Dioxide (as SO ₂)	µg/m ³	3.8	IS 5182: Part 02	80
6.	Suspended Particulate Matter (PM ₁₀)	µg/m ³	30.0	IS: 5182 Part 4	100
7.	Wind Speed	m/s	0.2	Instrument Method	NA
8.	Wind Direction	—	N-S	Instrument Method	NA

<--- End of Report --->

Verified By : <i>P. Ret.</i>	For ENVIRO CARE INDIA PRIVATE LIMITED (Laboratory Division) Authorized Signatory
Remarks : --	

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TEST REPORT

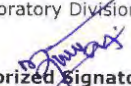
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AMBIENT AIR QUALITY SURVEY

Report No :	ECI-NN-AAQ-175/10/2020	Report Date :	19.10.2020
Customer Name & Address	M/s. P&C Projects (P) Ltd., WTP Project Site Pannaipatti, Viralipatti, Nilakottai Taluk, Battalagundu, Dindigul Dist.		
Customer Reference :	IWO Dt:14/10/2020	Sample Reference No :	ECI-NN-AAQ-175/10/2020
Sample Drawn By :	ECI	Sample Received On :	15.10.2020
Sample Collected Date :	14.10.2020	Test Commenced On :	15.10.2020
Qty of Sample Received :	Filter Paper & 25ml Solution	Test Completed On :	19.10.2020
Sample Description :	Ambient Air	Sampling Method :	IS 5182:P14
Latitude :	10.0892875 N	Longitude :	77.7296493 E
Sample Mark:	Water Treatment Plant- Sandiyur (Main Gate-Landscape Area)		

S.No	PARAMETERS	UNITS	RESULTS	TEST METHOD	Permissible limits of NAAQs (Industrial, Residential)
1.	Carbon Monoxide (as CO)	mg/m ³	< 1.2	ECI-SOP-SAM-08	2.0
2.	Lead (as Pb)	µg/m ³	< 0.1	IS 5182:Part 22	1.0
3.	Nitrogen dioxide (as NO ₂)	µg/m ³	10.2	IS 5182:Part 06	80
4.	Respirable Particulate Matter (PM _{2.5})	µg/m ³	15.4	IS 5182:Part 23	60
5.	Sulphur Dioxide (as SO ₂)	µg/m ³	4.8	IS 5182:Part 02	80
6.	Suspended Particulate Matter (PM ₁₀)	µg/m ³	31.0	IS:5182 Part 4	100
7.	Wind Speed	m/s	0.3	Instrument Method	NA
8.	Wind Direction	--	N-S	Instrument Method	NA

<--- End of Report --->

Verified By : <i>P. M. J.</i> Remarks : --	For ENVIRO CARE INDIA PRIVATE LIMITED (Laboratory Division)  Authorized Signatory
---	---

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TEST REPORT
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 Recognized by BIS as per IS : 14543 : 2016
AMBIENT AIR QUALITY SURVEY

Report No :	ECI-NN-AAQ-176/10/2020	Report Date :	19.10.2020
Customer Name & Address	M/s. P&C Projects (P) Ltd., WTP Project Site Pannaipatti, Viralipatti, Nilakottai Taluk, Battalagundu, Dindigul Dist.		
Customer Reference :	IWO Dt:14/10/2020	Sample Reference No :	ECI-NN-AAQ-176/10/2020
Sample Drawn By :	ECI	Sample Received On :	15.10.2020
Sample Collected Date :	14.10.2020	Test Commenced On :	15.10.2020
Qty of Sample Received :	Filter Paper & 25ml Solution	Test Completed On :	19.10.2020
Sample Description :	Ambient Air	Sampling Method :	IS 5182 P14
Latitude :	10.0892875 N	Longitude :	77.7296493 E
Sample Mark:	Existing WTP Plant-Near Clarifloculator & Chlorine Plant		

S.No	PARAMETERS	UNITS	RESULTS	TEST METHOD	Permissible limits of NAAQs (Industrial, Residential)
1	Carbon Monoxide (as CO)	mg/m ³	< 1.2	ECI-SOP-SAM-08	2.0
2	Lead (as Pb)	µg/m ³	< 0.1	IS 5182:Part 22	1.0
3	Nitrogen dioxide (as NO ₂)	µg/m ³	9.6	IS 5182:Part 06	80
4	Respirable Particulate Matter (PM _{2.5})	µg/m ³	17.6	IS 5182:Part 23	60
5	Sulphur Dioxide (as SO ₂)	µg/m ³	4.1	IS 5182:Part 02	80
6	Suspended Particulate Matter (PM ₁₀)	µg/m ³	31.4	IS:5182 Part 4	100
7	Wind Speed	m/s	0.4	Instrument Method	NA
8	Wind Direction	---	N-S	Instrument Method	NA

<--- End of Report --->

Verified By : <i>P. M. J.</i>	For ENVIRO CARE INDIA PRIVATE LIMITED (Laboratory Division)  Authorized Signatory
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TEST REPORT
 Accredited by **NABL** (Chemical & Biological)
 Recognized by BIS as per IS : 14543 : 2016
NOISE MONITORING

Report No :	ECI-NN-NM-51/10/2020	Report Date :	19.10.2020
Customer Name & Address	M/s. P&C Projects (P) Ltd., WTP Project Site Pannaipatti, Viralipatti, Nilakottai Taluk, Battalagundu, Dindigul Dist.		
Customer Reference :	IWO Dt:14/10/2020	Sample Reference No :	ECI-NN-NM-51/10/2020
Sample Description :	Noise	Monitoring Date :	14.10.2020
Monitored By :	ECI	Data Received On :	15.10.2020

S.No	Locations	Latitude	Longitude	Day Time	
				Minimum	Maximum
1.	Proposed WTP Plant Site	10.1135325 N	77.7353723 E	53.8	54.5
2.	Eastern Side of WTP Mr. Subbiah Garden (600 m Eastern Side of WTP)	10.1135325 N	77.7353723 E	52.8	53.2
3.	Existing WTP Plant near Clariflocculator & Chlorine Plant	10.0892875 N	77.7296493 E	52.5	53.8

Unit	dB (A)
TNPCB Max. Permissible Limit for Day Time (Residential Area)	55.0
Reference Method	IS 9989

<--- End of Report --->

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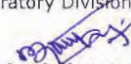
NOISE MONITORING

Report No :	ECI-NN-NM-52/10/2020	Report Date :	19.10.2020
Customer Name & Address	M/s. P&C Projects (P) Ltd., WTP Project Site Pannaipatti, Viralipatti, Nilakottai Taluk, Battalagundu, Dindigul Dist.		
Customer Reference :	IWO Dt:14/10/2020	Sample Reference No :	ECI-NN-NM-52/10/2020
Sample Description :	Noise	Monitoring Date :	14.10.2020
Monitored By :	ECI	Data Received On :	15.10.2020

S.No	Locations	Latitude	Longitude	Night time	
				Minimum	Maximum
1.	Existing WTP Site-Nearby Main Gate Landscape Area	10.0965 N	77.7324 E	42.5	43.9
2.	Proposed WTP Site	10.1135325 N	77.7353723 E	42.5	43.6
3.	Eastern Side of WTP Mr.Subbiah Gardern (600m Eastern Side of WTP)	10.1135325 N	77.7353723 E	41.3	42.9

Unit	dB (A)
TNPSB Max. Permissible Limit for Night Time (Residential Area)	45
Reference Method	IS 9989

<--- End of Report --->

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